01-40B CONTROL SYSTEM [WL-C, WE-C]

01

ENGINE CONTROL SYSTEM OUTLINE [WL-C, WE-C]

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Features

Improved emission gas purification	• Fuel injection control changed • EGR control (and intake shutter valve control) adopted

Specification

Item	Specification
IAT sensor No.2 (Inside MAF)	Thermistor
MAF sensor	Hot-wire
IAT sensor No.1	Thermistor
Boost sensor	Piezoelectric element
ECT sensor	Thermistor
CMP sensor	Hall element type
CKP sensor	Magnetic pickup
APP sensor	Potentiometer
EGR valve position sensor	Potentiometer
BARO sensor (built into PCM)	Piezoelectric element
Fuel temperature sensor	Thermistor
Fuel pressure sensor	Piezoelectric element
Neutral switch	ON/OFF
CPP switch	ON/OFF
Idle switch	ON/OFF

ENGINE CONTROL SYSTEM STRUCTURAL VIEW [WL-C, WE-C]

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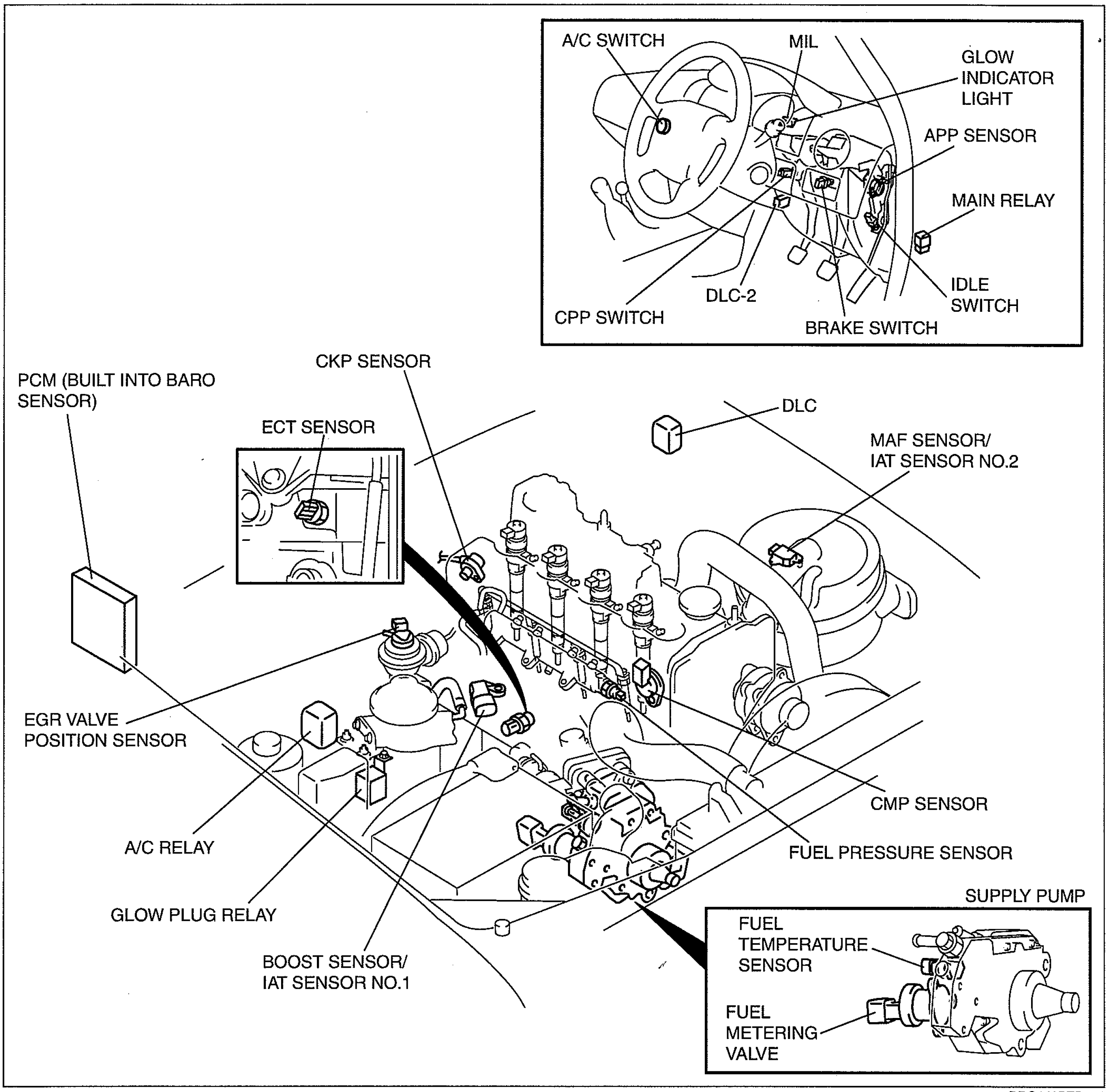

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ENGINE CONTROL SYSTEM DIAGRAM [WL-C, WE-C]

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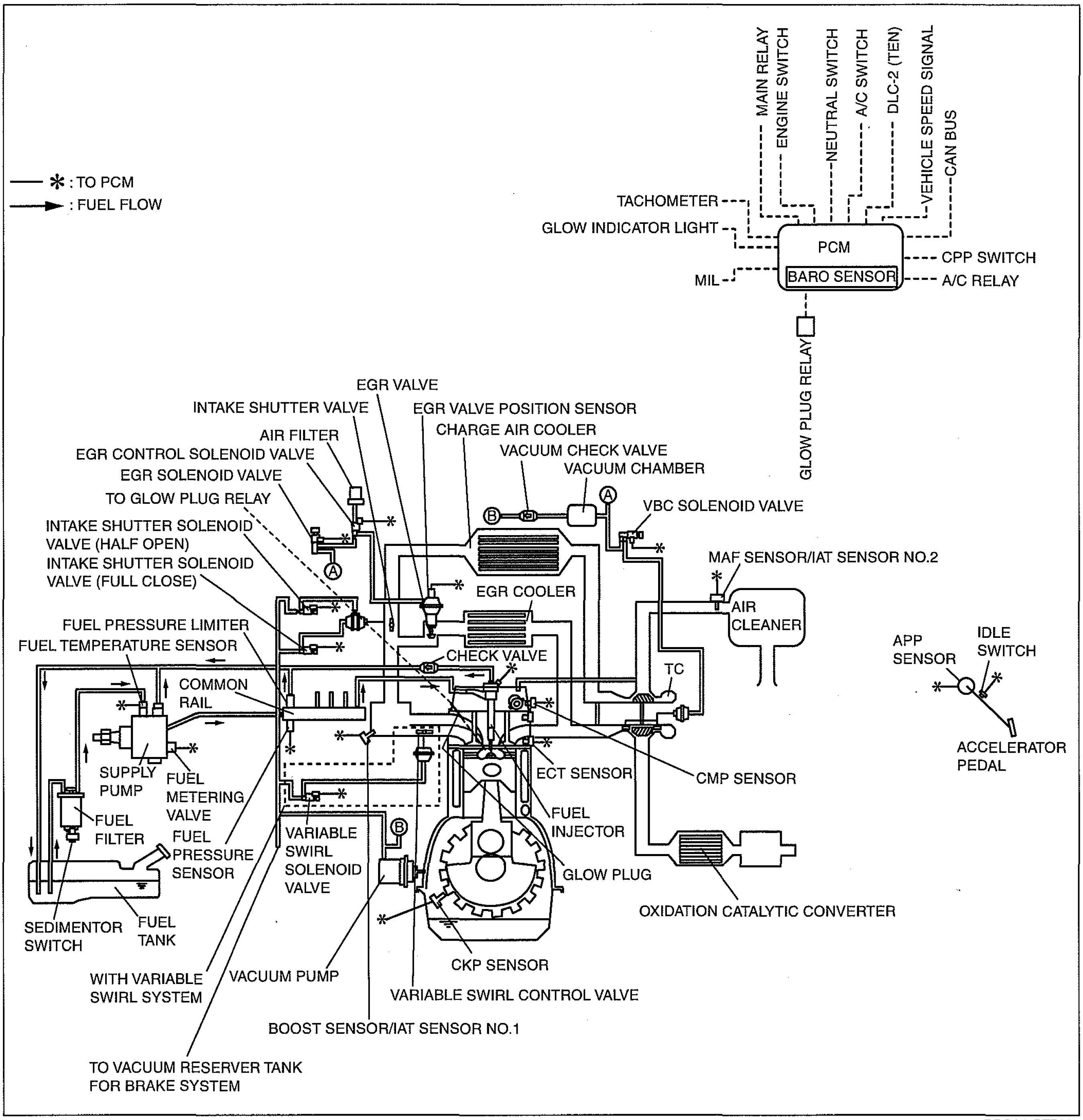

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ENGINE CONTROL SYSTEM WIRING DIAGRAM [WL-C, WE-C]

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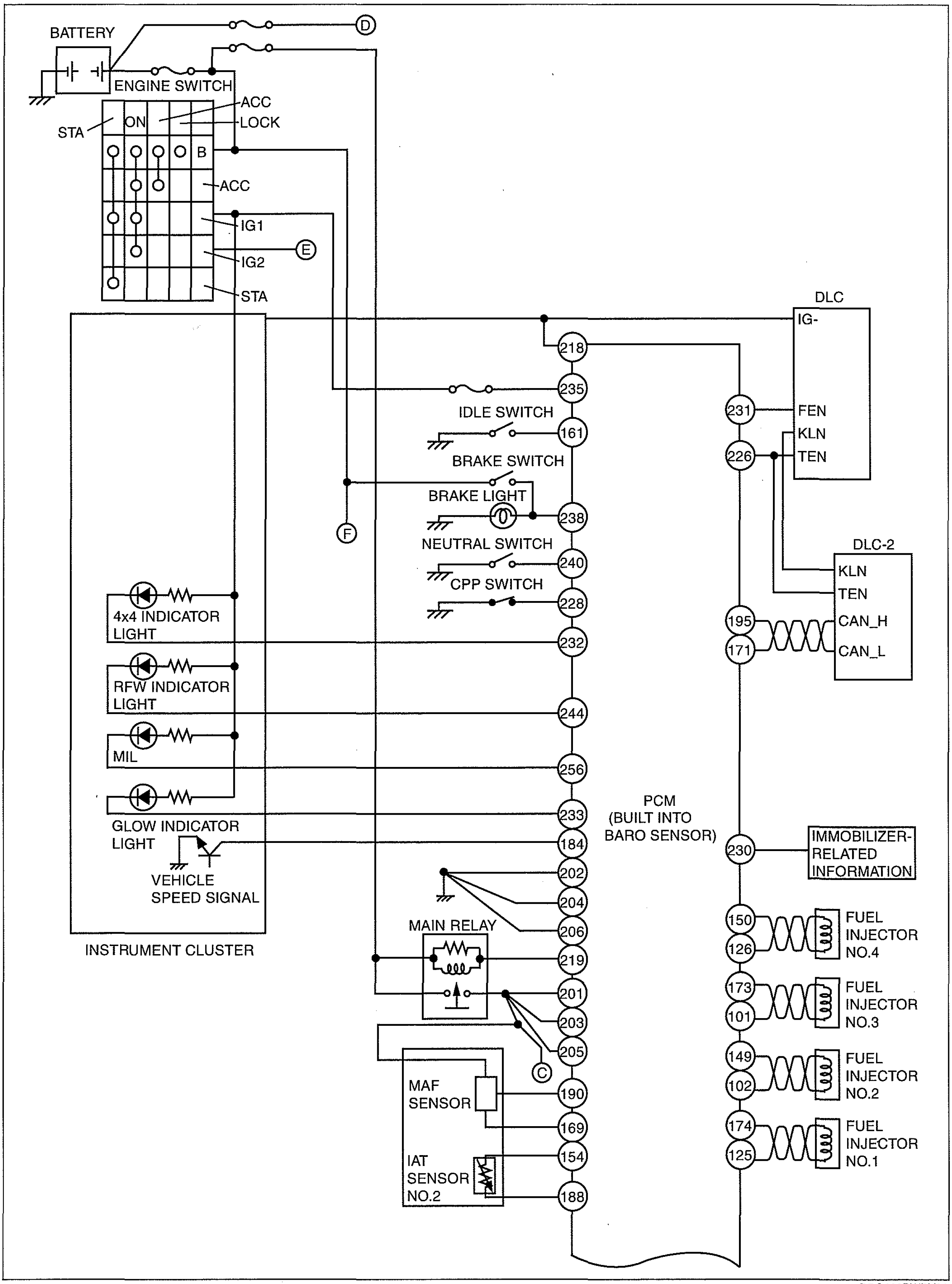
With Immobilizer System



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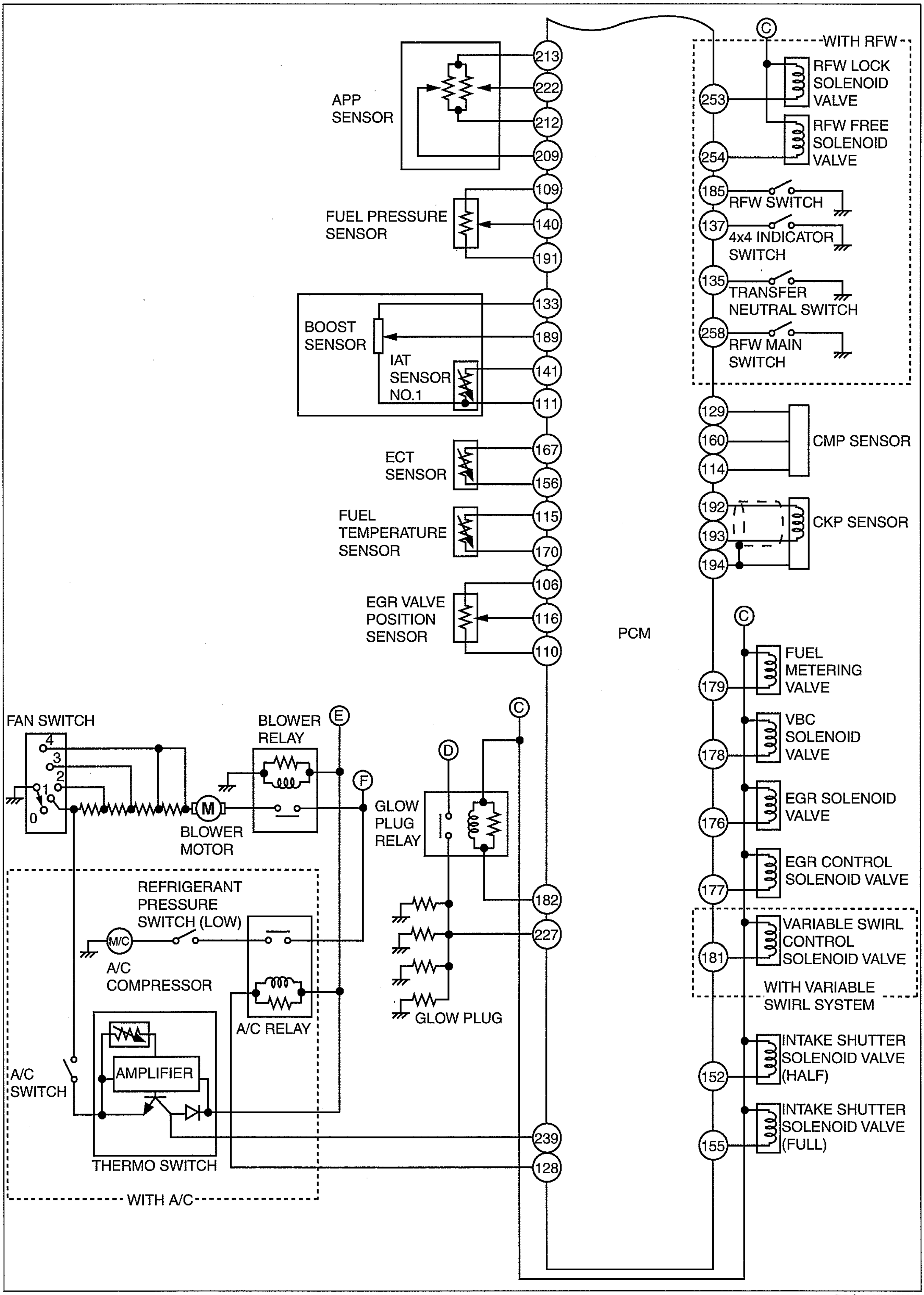

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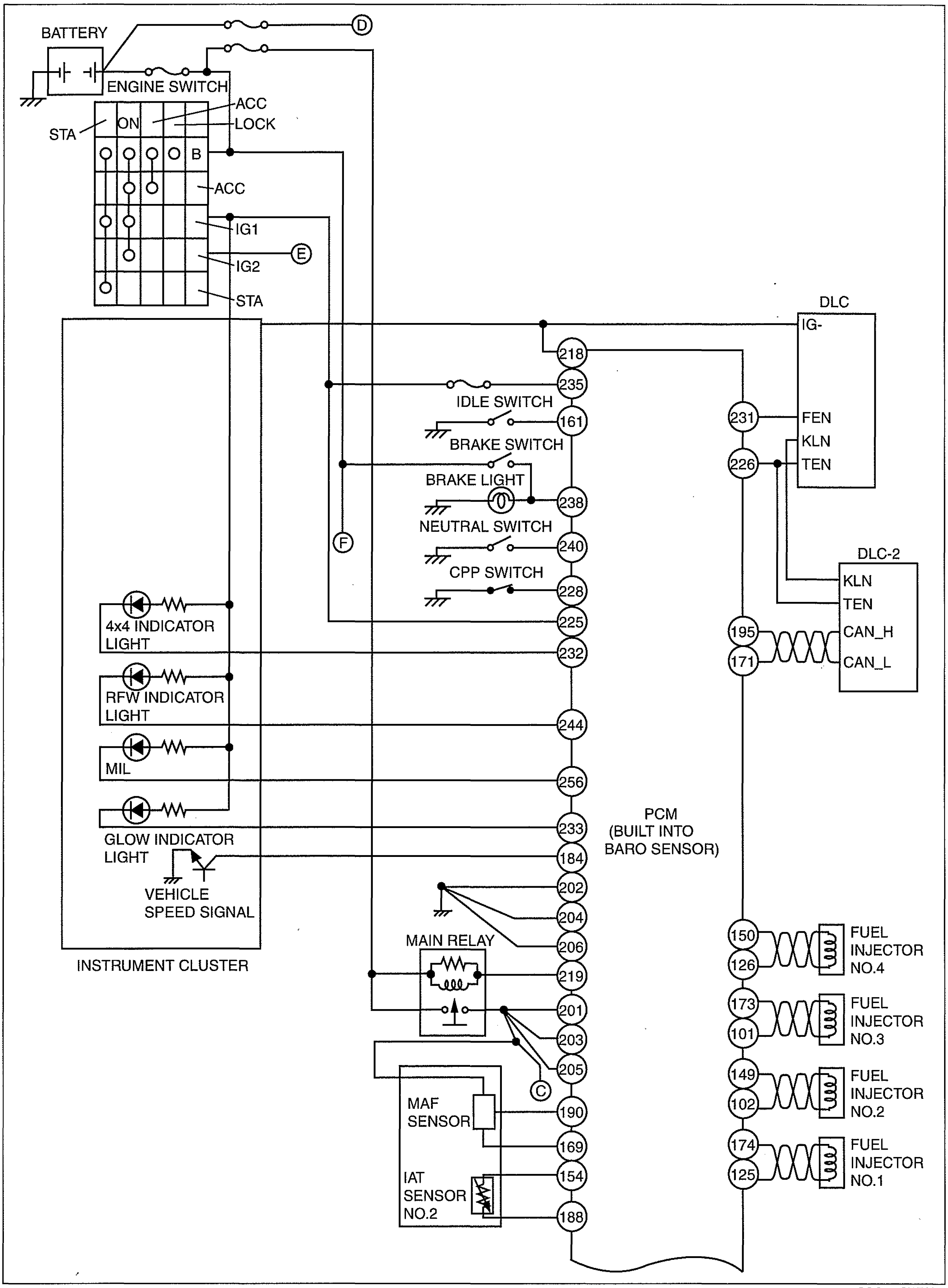
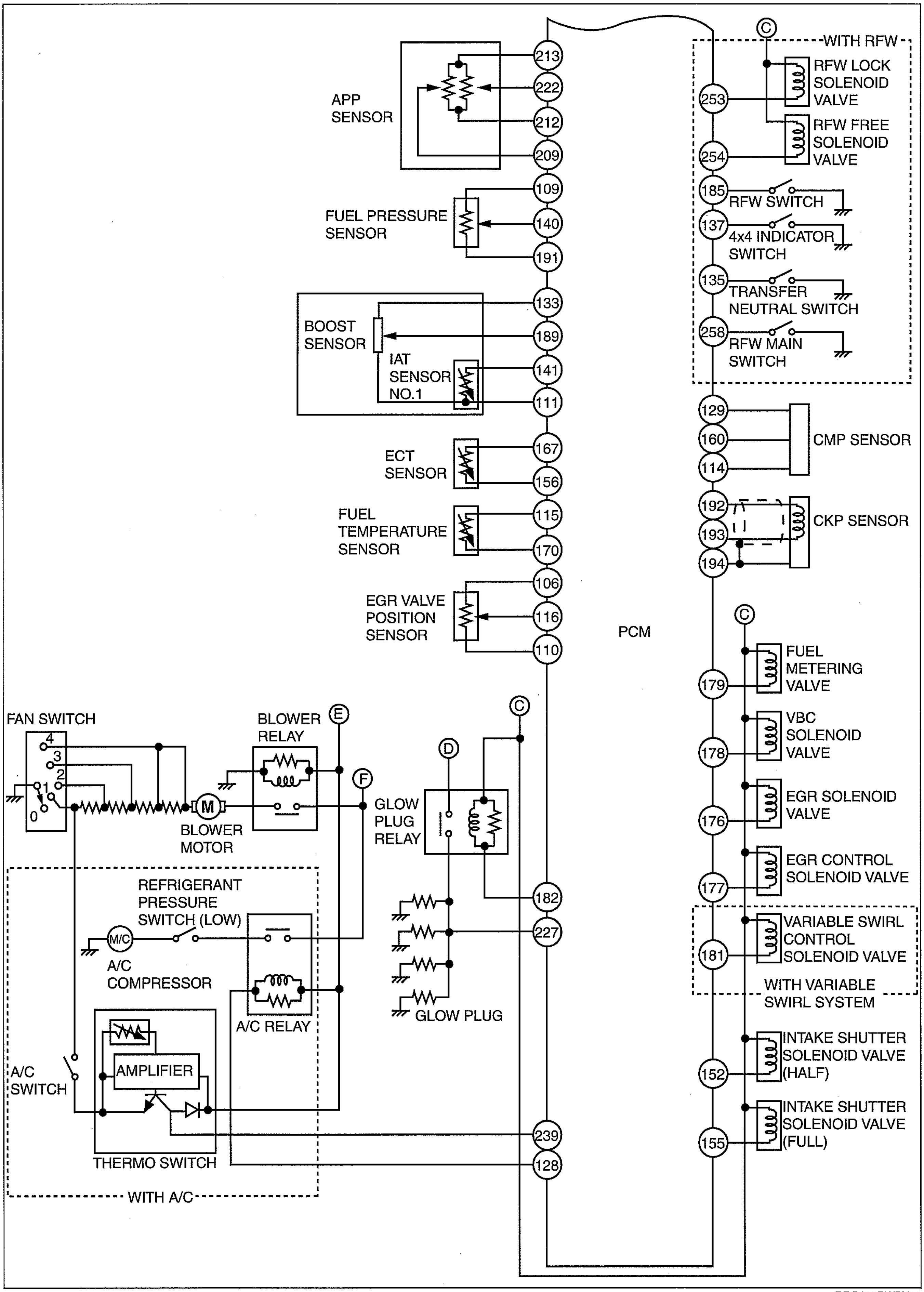
Without Immobilizer System





WITH RFW
RFW LOCK SOLENOID VALVE
RFW FREE SOLENOID VALVE
RFW SWITCH
4x4 INDICATOR SWITCH
TRANSFER NEUTRAL SWITCH
RFW MAIN SWITCH
APP SENSOR
FUEL PRESSURE SENSOR
BOOST SENSOR
IAT SENSOR NO.1
ECT SENSOR
FUEL TEMPERATURE SENSOR
EGR VALVE POSITION SENSOR
CMP SENSOR
CKP SENSOR
PCM
FUEL METERING VALVE
VBC SOLENOID VALVE
EGR SOLENOID VALVE
EGR CONTROL SOLENOID VALVE
VARIABLE SWIRL CONTROL SOLENOID VALVE
WITH VARIABLE SWIRL SYSTEM
INTAKE SHUTTER SOLENOID VALVE (HALF)
INTAKE SHUTTER SOLENOID VALVE (FULL)
FAN SWITCH
BLOWER RELAY
BLOWER MOTOR
GLOW PLUG RELAY
GLOW PLUG
REFRIGERANT PRESSURE SWITCH (LOW)
A/C COMPRESSOR
A/C RELAY
A/C SWITCH
AMPLIFIER
THERMO SWITCH
WITH A/C
213 222 212 209 109 140 191 133 189 141 111 167 156 115 170 106 116 110 182 227 239 128
253 254 185 137 135 258 129 160 114 192 193 194 179 178 176 177 181 152 155
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ENGINE CONTROL SYSTEM BLOCK DIAGRAM [WL-C, WE-C]

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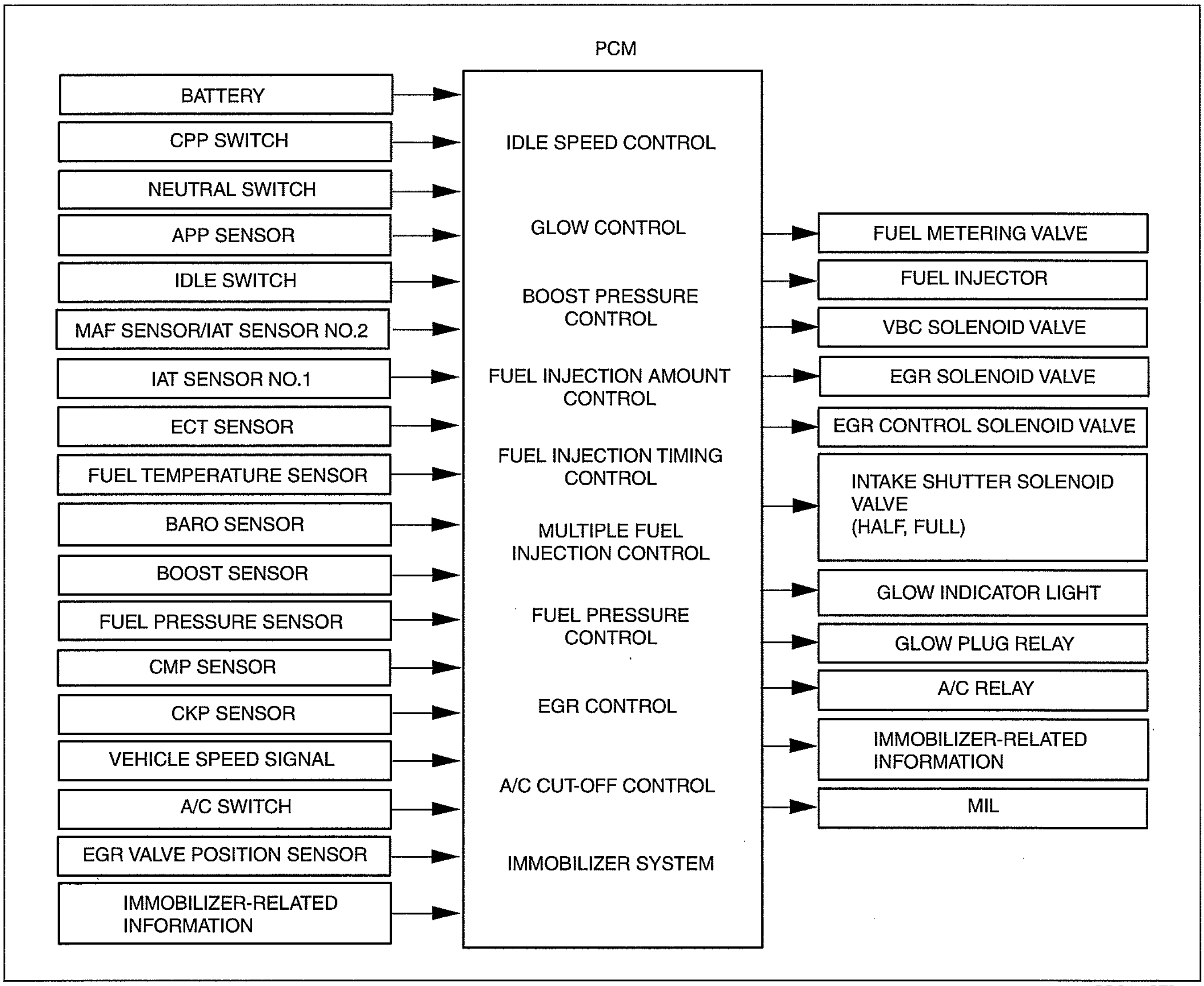

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ENGINE CONTROL SYSTEM RELATION CHART [WL-C, WE-C]

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×: Applicable

Item	Idle speed control	Glow control	Boost pressure control	Fuel injection amount control	Fuel injection timing control	Multiple fuel injection control	Fuel pressure control	EGR control	A/C cut-off control	Immobilizer system
Input device										
Battery			×							
CPP switch	×			×	×	×		×		
Neutral switch	×			×	×	×		×		
APP sensor	×		×	×	×	×	×	×	×	
Idle switch				×				×		
MAF sensor/IAT sensor No.2				×	×	×	×	×		
IAT sensor No.1				×	×					
ECT sensor	×	×	×	×	×	×	×	×	×	
Fuel temperature sensor				×	×		×			
BARO sensor			×	×	×	×	×	×		
Boost sensor			×							
Fuel pressure sensor				×			×	×		
CMP sensor	×			×	×	×				
CKP sensor	×	×	×	×	×	×	×	×	×	
Vehicle speed signal	×			×				×	×	
A/C switch	×									
EGR valve position sensor								×		
Immobilizer-related information										×
Output device										
Fuel metering valve							×			×
Fuel injector	×			×	×	×				×
VBC solenoid valve			×							
EGR solenoid valve								×		
EGR control solenoid valve								×		
Intake shutter solenoid valve (half, full)								×		
Glow indicator light		×								
Glow plug relay		×								
A/C relay									×	
Immobilizer-related information										×

IDLE SPEED CONTROL OUTLINE [WL-C, WE-C]

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- The PCM feedback controls the idle speed by calculating the fuel injection amount so that the idle speed is at the target idle speed corresponding to the driving conditions.

IDLE SPEED CONTROL BLOCK DIAGRAM [WL-C, WE-C]

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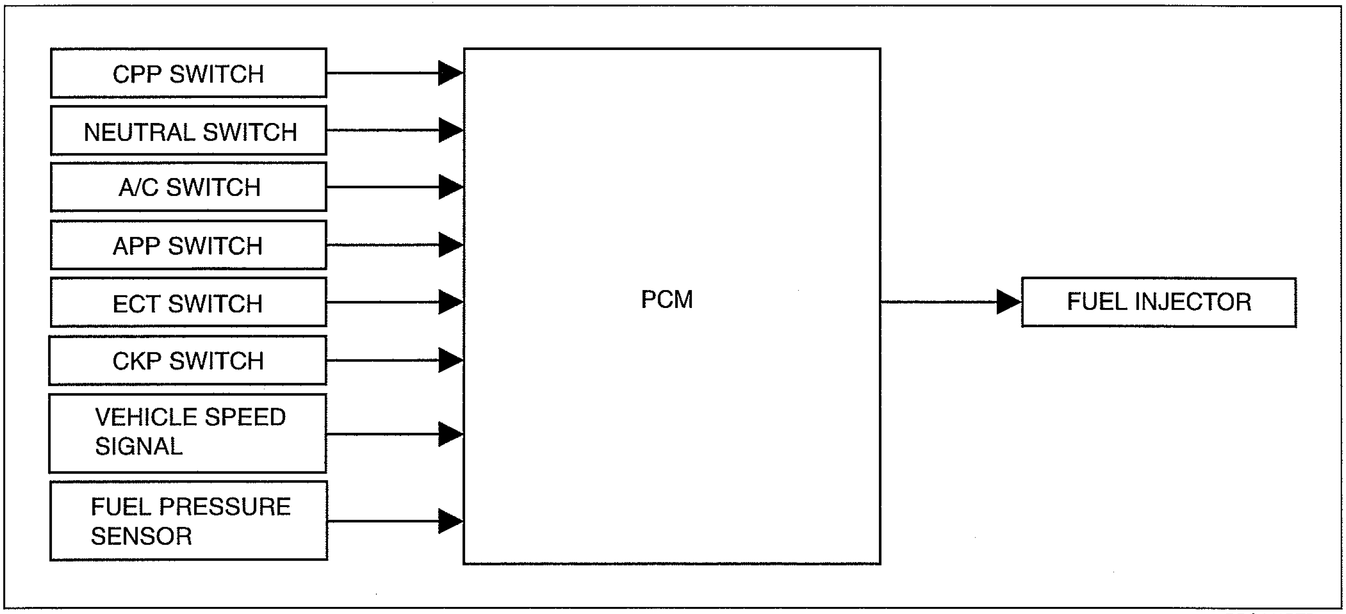

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IDLE SPEED CONTROL OPERATION [WL-C, WE-C]

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Feedback Control

- If there is a difference between the target idle speed and the actual idle speed, the PCM controls the fuel injection amount so that the actual idle speed corresponds to the target idle speed.

Target idle speed
No load: 720 rpm
A/C ON: 750 rpm

Engine Speed Control while Warming Up

- Controls engine speed based on the input signal from the engine coolant temperature sensor (below 0 °C {32 °F}) so that the engine speed while warming up corresponds to the target engine speed.

Engine speed while warming up control
720—1,000 rpm

Engine Speed Fluctuation Prevention Control for Each Cylinder

- The PCM detects the engine speed fluctuation while idling and corrects the fuel injection amount per each cylinder. Due to this, the variance in fuel injection amount caused by ununiformity of the cylinders (such as fuel injectors) and engine vibration while idling are reduced.

GLOW CONTROL OUTLINE [WL-C, WE-C]

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- Heating of the glow plug is controlled by the PCM through the glow plug relay to improve engine startability.
- Energization time to the glow plug is determined according to the engine coolant temperature and engine starting conditions.

GLOW CONTROL BLOCK DIAGRAM [WL-C, WE-C]

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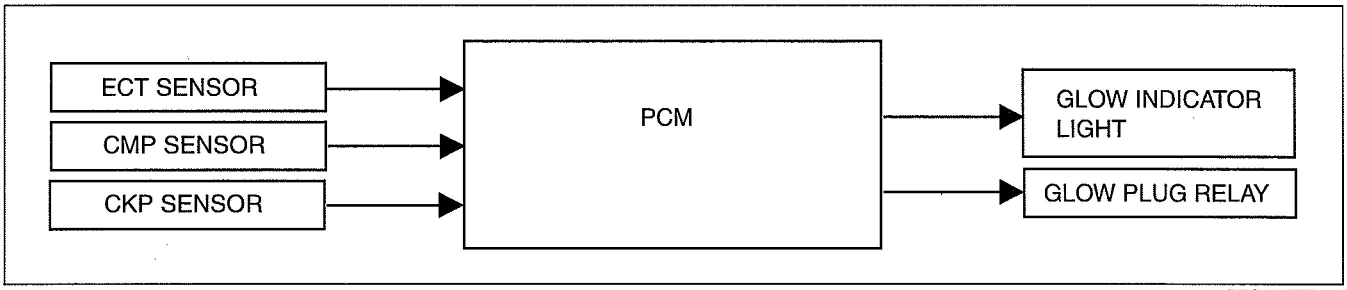

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GLOW CONTROL OPERATION [WL-C, WE-C]

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- The glow plug relay operates under the following conditions:
  — During preheating timer operation
  — During engine cranking
  — While outer glow is operating

Preheating Timer

- The PCM operates the glow plug relay for several seconds after the engine switch is turned to the ON position according to the engine coolant temperature. The higher the engine coolant temperature, the shorter the period of the timer operation.
- The glow indicator light illuminates during the preheating timer operation to inform the driver that the glow system is operating.

Glow Continuation Timer

- The PCM operates the glow plug relay for 120 s (max.). after the engine has been started with the engine coolant temperature lower than 32 °C {89.6 °F}. When the engine coolant temperature reaches 32 °C {89.6 °F} or more, the glow continuation timer is cancelled.

BOOST PRESSURE CONTROL OUTLINE [WL-C, WE-C]

dcf014000000t34

- The PCM duty-controls the VBC solenoid valve.
- By changing the opening angle of the guide blades (variable vanes) around the turbine impellers according to the engine driving conditions, turbocharger response, vehicle acceleration performance, and fuel consumption have been improved in every driving range.

BOOST PRESSURE CONTROL BLOCK DIAGRAM [WL-C, WE-C]

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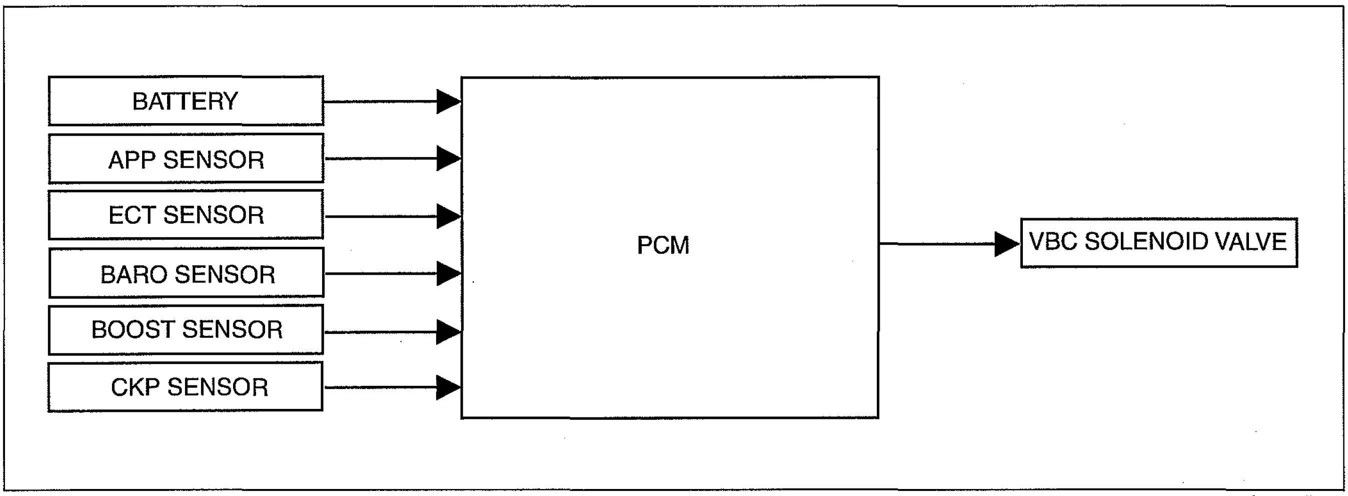

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BOOST PRESSURE CONTROL OPERATION [WL-C, WE-C]

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- The VBC solenoid valve is controlled by the variable boost control signal (duty signal) from the PCM. The longer the valve is ON, the higher the boost pressure becomes.
- The PCM performs feedback control by determining the period that the VBC solenoid valve is ON based on the engine speed and the fuel injection amount so that the optimum boost pressure can be obtained.
- Under driving conditions where more boost pressure is needed, the PCM sends the variable boost control signal to extend the ON period of the VBC solenoid valve, moving the guide blades in the turbocharger in the closing direction.
- Under driving conditions where less boost pressure is needed, the PCM sends the variable boost control signal to shorten the ON period of the VBC solenoid valve, moving the guide blades in the turbocharger in the opening direction.

FUEL INJECTION AMOUNT CONTROL OUTLINE [WL-C, WE-C]

dcf014000000t37

- The fuel injection amount is controlled by the opening of the nozzle in the fuel injector based on the signal from the PCM.
- The PCM corrects the variation in the fuel injection amount of each cylinder by inputting the fuel injector calibration code (or fuel injector calibration bar code) into the PCM.

FUEL INJECTION AMOUNT CONTROL BLOCK DIAGRAM [WL-C, WE-C]

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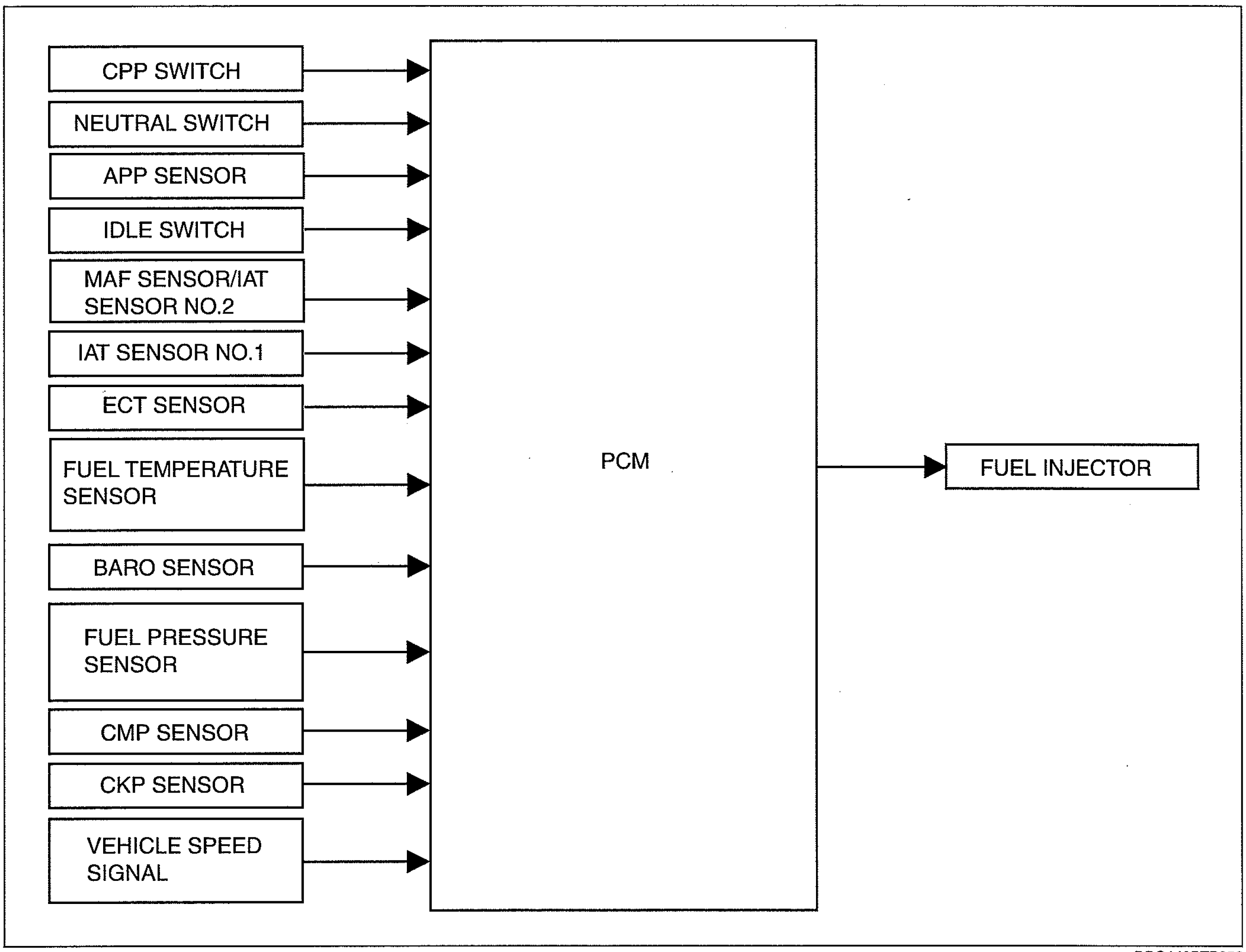

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FUEL INJECTION AMOUNT CONTROL OPERATION [WL-C, WE-C]

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Basic Injection Amount

Operation
- Fuel injection amount is controlled based on the fuel injector nozzle opening period.

Injection amount calculation
- The PCM calculates the optimum fuel injection amount according to engine operation conditions.
- The following two amount values are compared and the one of less amount is selected for the final injection amount.

Basic injection amount
- The logically necessary injection amount is calculated based on the accelerator opening angle and engine speed.

Maximum injection amount
- The maximum injection amount is calculated by correcting the basic injection amount based on driving conditions such as mass air flow, intake air temperature, and barometric pressure, intake air temperature and engine coolant temperature.

FUEL INJECTION TIMING CONTROL OUTLINE [WL-C, WE-C]

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- The PCM determines and controls the optimum fuel injection timing according to engine operation conditions based on the signals from each input sensor.

FUEL INJECTION TIMING CONTROL BLOCK DIAGRAM [WL-C, WE-C]

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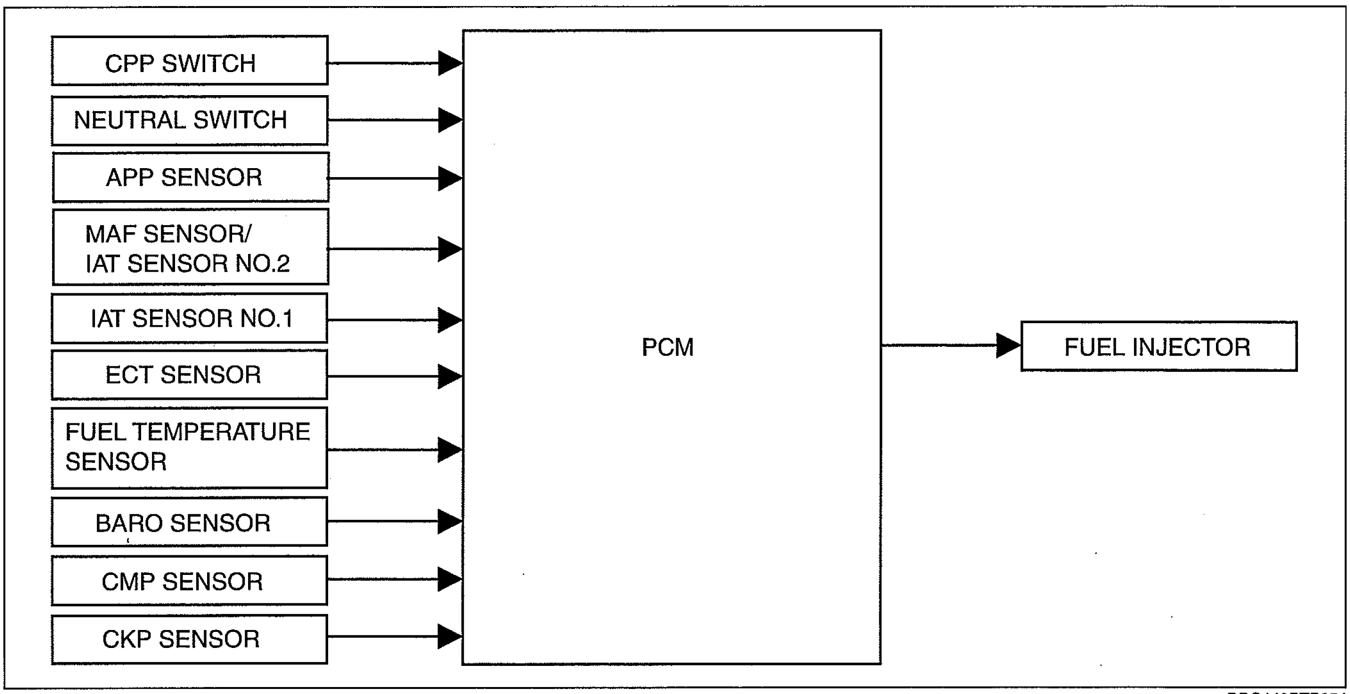

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FUEL INJECTION TIMING CONTROL OPERATION [WL-C, WE-C]

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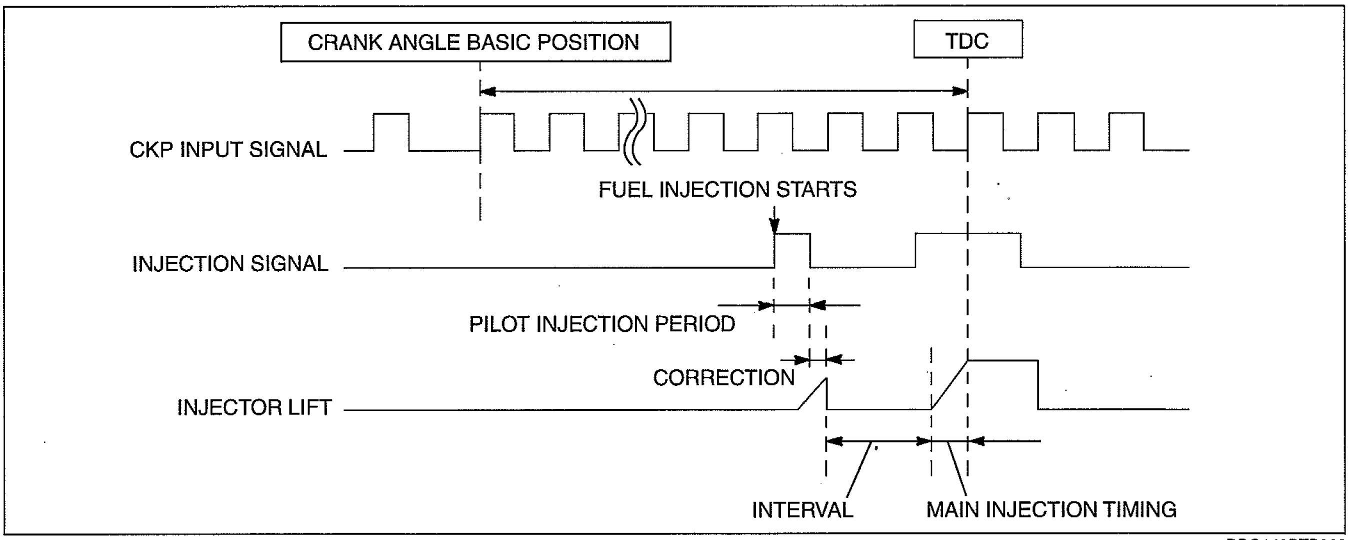
- Fuel injection start timing is determined based on the fuel injection amount and crank angle signals input from the crankshaft position sensor.



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Determination of Fuel Injection Timing

- The PCM calculates the optimum fuel injection timing based on the predetermined target fuel injection timing and input signals from each sensor.

Target Fuel Injection Timing

- Target fuel injection timing is calculated based on engine speed and fuel injection amount.

Injection Timing Correction

- Actual fuel injection timing is corrected based on the input signals from each sensor such as intake air temperature, engine coolant temperature, and barometric pressure.

MULTIPLE FUEL INJECTION CONTROL OUTLINE [WL-C, WE-C]

dcf014000000t43

- Pilot injection: To reduce noise and NOx, the PCM performs multiple fuel injections before the main injection according to the vehicle driving conditions.

MULTIPLE FUEL INJECTION CONTROL BLOCK DIAGRAM [WL-C, WE-C]

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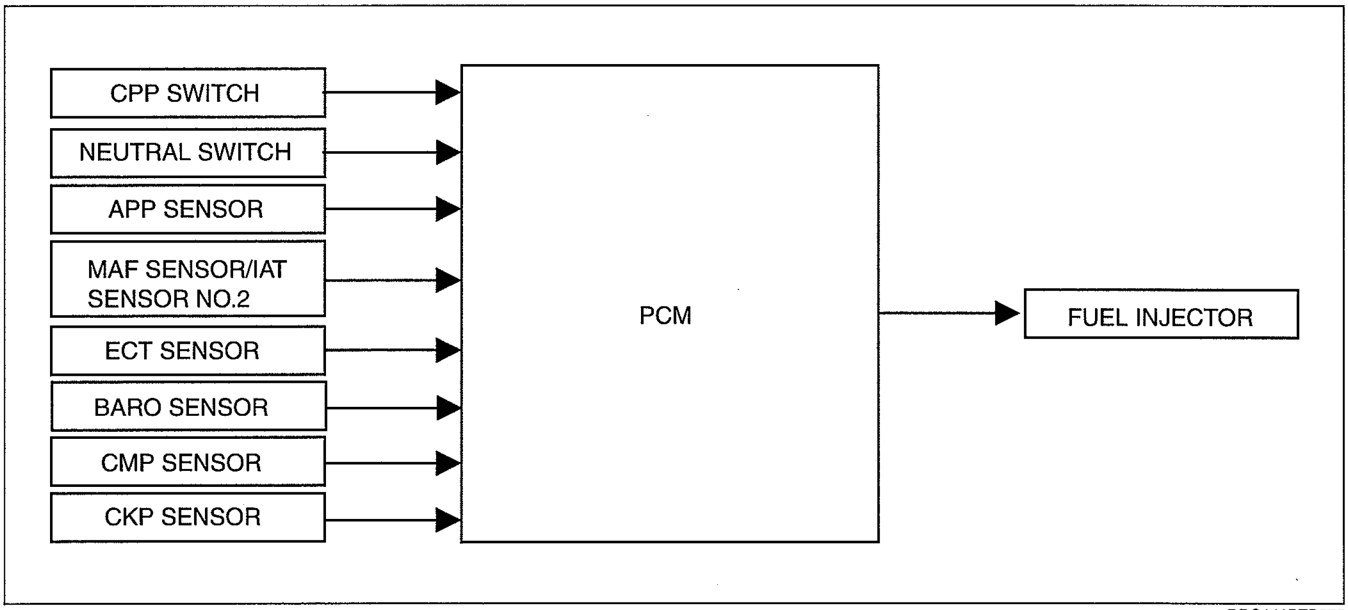

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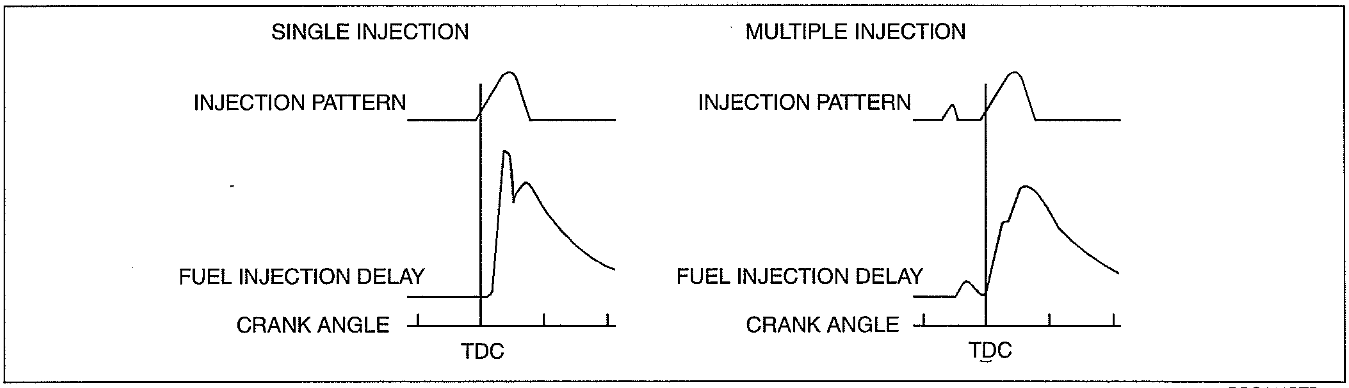
MULTIPLE FUEL INJECTION CONTROL OPERATION [WL-C, WE-C]

dcf014000000t45

Function

- In single injection, ignition delay becomes longer and much fuel is injected before ignition since the ignition has started. Therefore, the fuel is combusted rapidly at a high temperature and a high pressure, resulting in increased noise and NOx.
- To prevent this, the PCM performs multiple fuel injection. Due to this, the fuel is combusted slowly and the occurrence of noise and NOx is suppressed.



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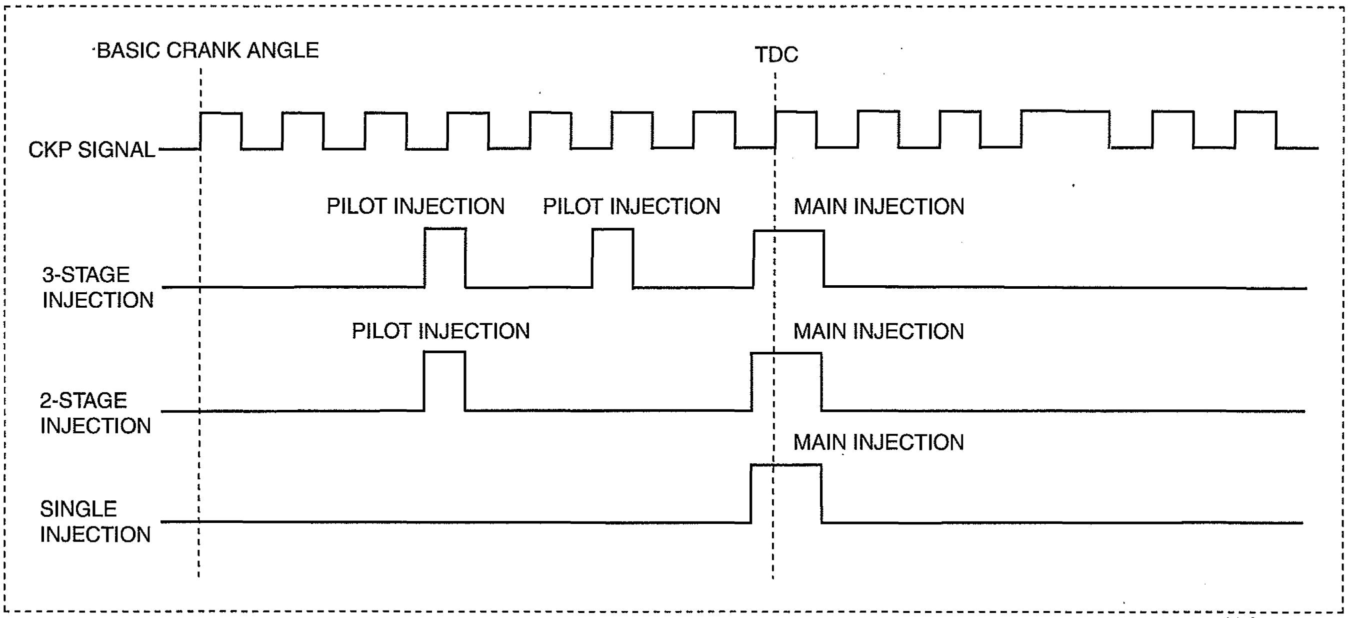
Operation

- Injection patterns of the multistage injection control according to the combination of injection times are shown in the figure. The PCM selects the optimum injection pattern based on the vehicle conditions and performs injection for 3 times at maximum.
- While in a driving condition at low engine speed and low engine load, the fuel injection time is increased up to 3 times to reduce knocking noise which occurs during combustion.
- While in a driving condition at high engine speed and high engine load, the fuel injection time is reduced to one time at minimum to improve output and fuel economy.



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Operation condition

- The PCM determines the number of times fuel injection occurs based on the engine speed, fuel injection amount, and signals from each sensor.
- When the engine is started, the PCM calculates the number of times fuel injection occurs based on the engine coolant temperature, engine speed and fuel injection amount.

FUEL PRESSURE CONTROL OUTLINE [WL-C, WE-C]

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- The PCM performs the feedback control of the fuel pressure in the common rail to gain optimum fuel injection pressure according to engine operation conditions.
- Because the fuel pressure can be controlled regardless of engine operation conditions, high pressure fuel injection even with low engine speed is possible. Due to this, generation of NOx and particulate matter can be reduced.

FUEL PRESSURE CONTROL BLOCK DIAGRAM [WL-C, WE-C]

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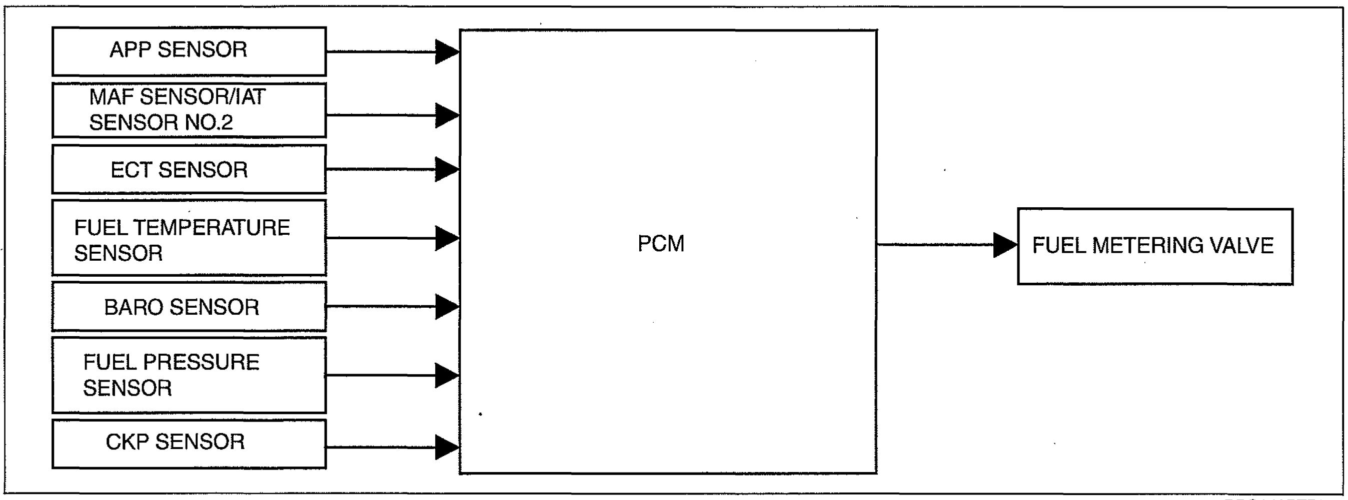

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FUEL PRESSURE CONTROL OPERATION [WL-C, WE-C]

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- The PCM calculates the target fuel pressure based on the engine speed and fuel injection amount.
- The PCM operates the fuel metering valve which is installed to the supply pump to adjust the fuel pressure in the common rail to the target fuel pressure.
- By controlling the amount of pumped fuel from the supply pump with the fuel metering valve, the fuel pressure in the common rail is controlled and a constant, optimum injection pressure has been realized.

EGR CONTROL OUTLINE [WL-C, WE-C]

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- The PCM controls duty-cycle type solenoid valves and an ON/OFF type solenoid valve to control the EGR valve in order to obtain the optimum EGR amount for the actual engine operating condition. Due to this, emission performance and driveability has been improved.
- When introducing the exhaust gas during low engine speed, the PCM controls the intake shutter solenoid valve (half) to control the intake shutter valve opening angle. Thus the exhaust gas is channelled into the combustion chamber efficiently. Due to this, emission performance and driveability has been improved. (WE-C)
- As the MAF sensor deteriorates and an error margin is created between the target and the actual intake air amount, the PCM conducts learning and correction.

EGR CONTROL BLOCK DIAGRAM [WL-C, WE-C]

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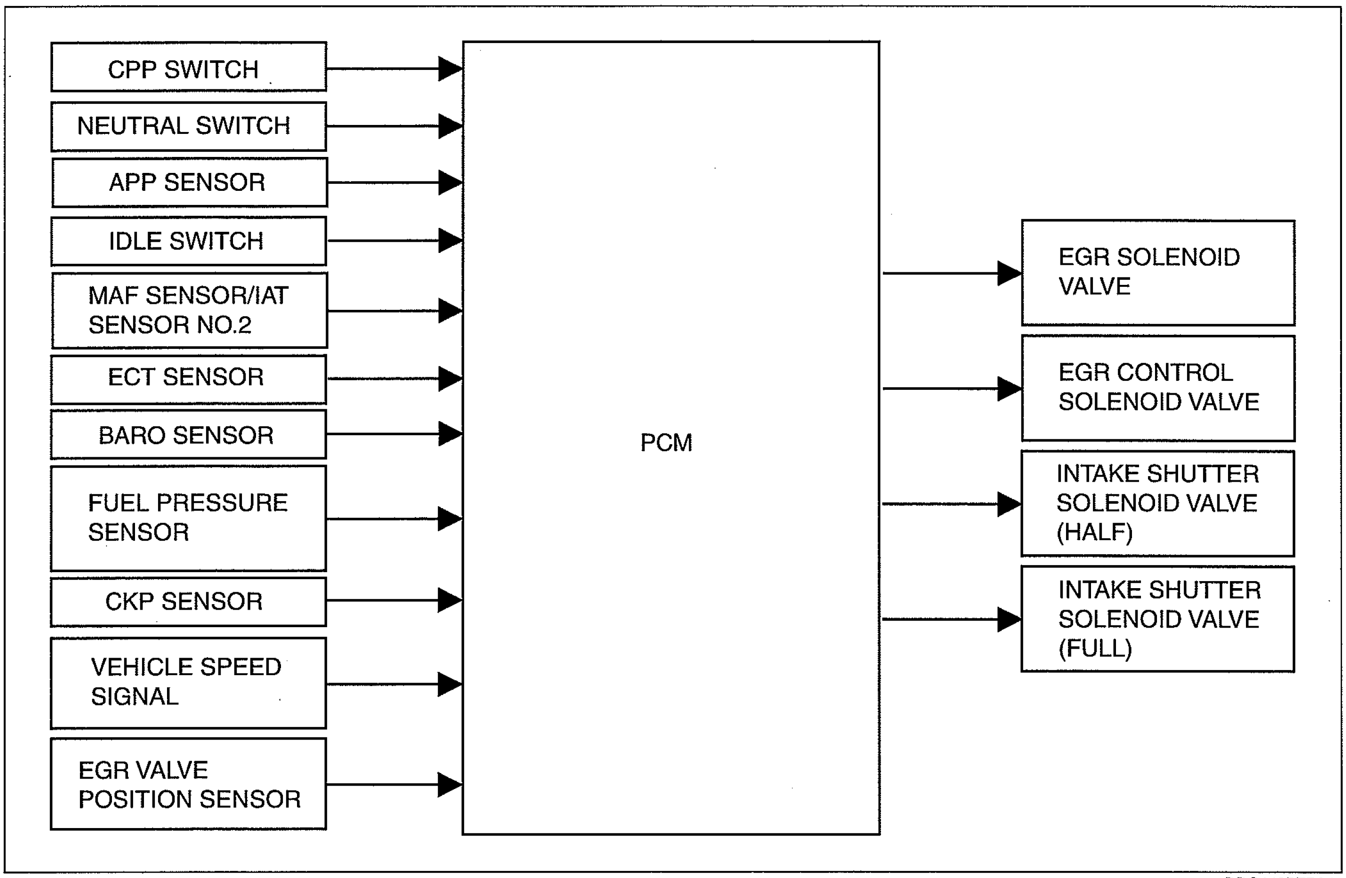

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EGR CONTROL OPERATION [WL-C, WE-C]

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EGR Valve Operation

- To improve emission performance, target EGR valve position is decided based on intake air flow, engine speed, engine coolant temperature and fuel injection amount.
- The flow amount of the EGR is calculated from the difference between the following amounts.
  — Target intake air amount that is mainly calculated from engine speed and fuel injection amount
  — Actual intake air amount that is mainly detected by MAF sensor/IAT sensor No.2
- The PCM controls the vacuum applied to the EGR valve by opening and closing a two-duty solenoid valve.

Intake Shutter Valve Operation

Close condition

- The PCM turns on the intake shutter solenoid valve (half) and the intake shutter solenoid valve (full) in order to close the intake shutter valve when the engine switch is at OFF.

Halfway open condition (WE-C)

- The PCM turns the intake shutter solenoid valve (half) on when all of the following conditions are met.
  - Engine coolant temperature is above threshold level.
  - Intake air temperature is threshold level.
  - Barometric pressure is above threshold level.
  - Engine speed is below threshold level.
  - Vehicle speed is below threshold level.
  - Fuel injection amount is below threshold level.

Full open condition

- For conditions other than closed and halfway open, the PCM turns off the intake shutter solenoid valve (half) and the intake shutter solenoid valve (full) in order to open the intake shutter valve.

MAF Learning

- After learning is initiated, the PCM calculates the actual intake air amount for the complete learning period.
- The PCM calculates the deviation between the target and actual intake air amount, then, based on the deviation, derives the learning factor. The PCM stores the learning factor and uses it to correct the intake air amount. Due to this, the error margin of the MAF sensor is resolved. If the deviation factor is less than the specified amount, the PCM does not calculate nor update the learning factor.
- The PCM stores the learning factor until the next time the factor is updated. The factor is erased using the M-MDS.

Using the M-MDS for learning

- By using the M-MDS, the learning function can be freely controlled. Due to this, even if the PCM or MAF sensor/IAT sensor No.2 is replaced, the learning factor can be stored again.
- When using the M-MDS to control the MAF learning function, learning is performed at the following three engine speed levels.
  - 720 rpm
  - 1,850 rpm
  - 2,500 rpm

Without using the M-MDS for learning

- By shorting circuit the DLC terminal TEN five times within 5 s, the learning function can be freely controlled. Due to this, even if the PCM or MAF sensor/IAT sensor No.2 is replaced, the learning factor can be stored again.
- The glow indicator light illuminates while MAF learning is performed, and flashes five times after it is completed.

A/C CUT-OFF CONTROL OUTLINE [WL-C, WE-C]

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- The A/C relay is turned on and off according to engine operation conditions to improve drivability.

A/C CUT-OFF CONTROL BLOCK DIAGRAM [WL-C, WE-C]

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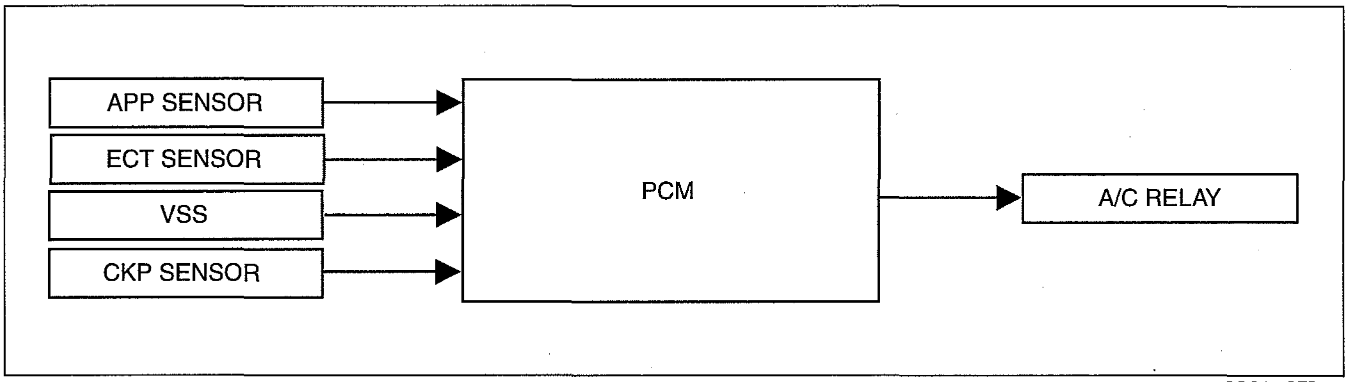

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A/C CUT-OFF CONTROL OPERATION [WL-C, WE-C]

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- When the accelerator opening angle is 87.5 % or more, the PCM cuts the power supply to the A/C relay for 3 s.
- When the engine coolant temperature is 113 °C {235 °F} or more, the PCM turns off the power supply to the A/C relay until the engine coolant temperature is less than 110 °C {230 °F}.

IMMOBILIZER SYSTEM OUTLINE [WL-C, WE-C]

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- While the immobilizer system is operating, the PCM cuts off fuel delivery and stops pressure feeding fuel to the common rail.

IMMOBILIZER SYSTEM BLOCK DIAGRAM [WL-C, WE-C]

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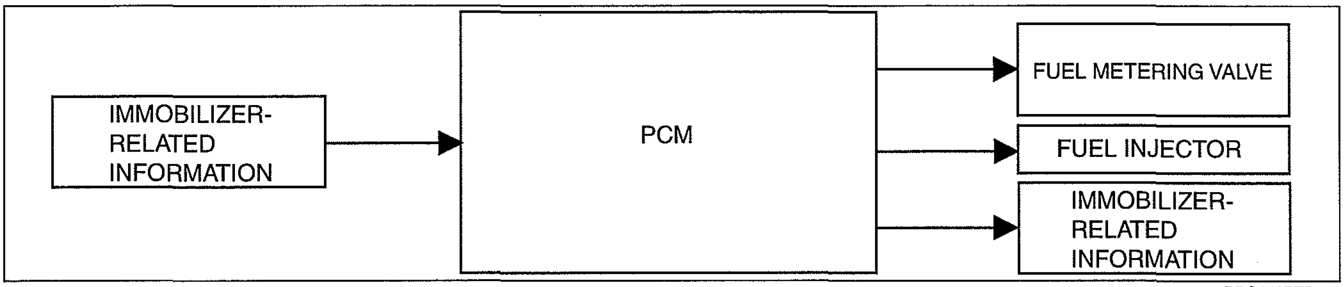

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IMMOBILIZER SYSTEM OPERATION [WL-C, WE-C]

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- When the immobilizer system is activated, the following controls are carried out.
  — Fuel injector: OFF (Set fuel quantity to zero.)
  — Fuel metering valve: ON (Set minimum common rail pressure.)

PCM FUNCTION [WL-C, WE-C]

dcf014018880t03

Function Table

- Control descriptions are as follows:

Function	Contents
Idle speed control	Performs feedback control of idle speed by calculating fuel injection amount so that idle speed is at target idle speed corresponding to driving conditions.
Glow control	Controls glow plug operation time via glow plug relay to improve engine startability.
Boost pressure control	Controls turbocharger guide blade angle according to engine operation conditions to improve turbocharger response, acceleration performance, and fuel economy.
Fuel injection amount control	Performs optimum fuel injection according to engine operation conditions by controlling signals to open/close fuel injector nozzle.
Fuel injection timing control	Controls optimum fuel injection timing according to engine operation conditions.
Multiple fuel injection control	Performs multiple fuel injection according to engine operation conditions to reduce noise and NOx.
Fuel pressure control	Adjusts to optimum fuel pressure according to engine operation conditions.
Intake shutter valve control	Adjusts intake shutter valve to optimum opening angle according to engine operation conditions.
EGR control	Controls EGR valve so that EGR volume is optimized according to engine operation conditions.
A/C cut-off control	Controls operation of A/C according to engine operation conditions to assure drivability.
Immobilizer system	Inhibits engine starting by cutting fuel injection and pressure feeding fuel to common rail when immobilizer system is activated.

PCM CONSTRUCTION [WL-C, WE-C]

dcf014018880t04

- A 154-pin (two block) PCM connector has been adopted.
- The PCM has a built into piezoelectric element BARO sensor.

MASS AIR FLOW (MAF) SENSOR FUNCTION [WL-C, WE-C]

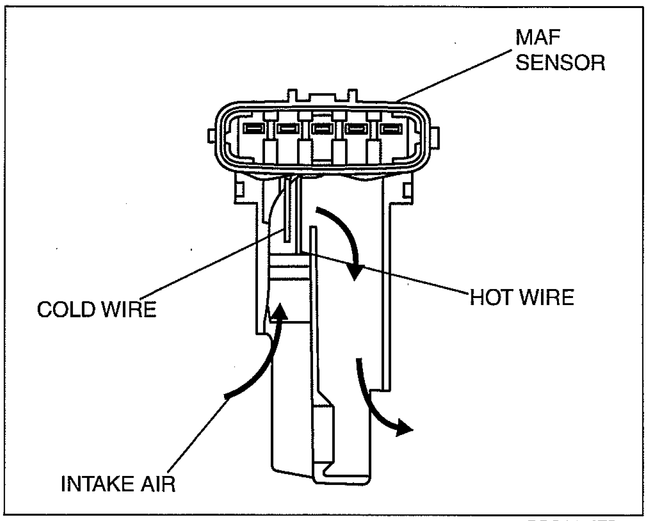
dcf014013215t03

- Detects the intake air amount (mass airflow amount).

MASS AIR FLOW (MAF) SENSOR CONSTRUCTION/OPERATION [WL-C, WE-C]

dcf014013215t04

- Installed to the air cleaner.
- With built-in IAT sensor No.2
- Converts the mass intake airflow amount to voltage.
- When the temperature of the metal decreases, the resistance decreases. Using this characteristic, the hot wire captures heat from the flow of intake air and converts the intake airflow amount to voltage.
- The cold wire converts intake air density to voltage from the ambient temperature of the cold wire, using the characteristic of air whereby the intake air density decreases due to the increase in intake air temperature.
- The voltages obtained by the hot wire (intake airflow amount) and the cold wire are compared and the electric potential becomes stable by supplying the difference in voltage to the transistor. The voltage supplied to the hot wire is output as the mass intake airflow amount.



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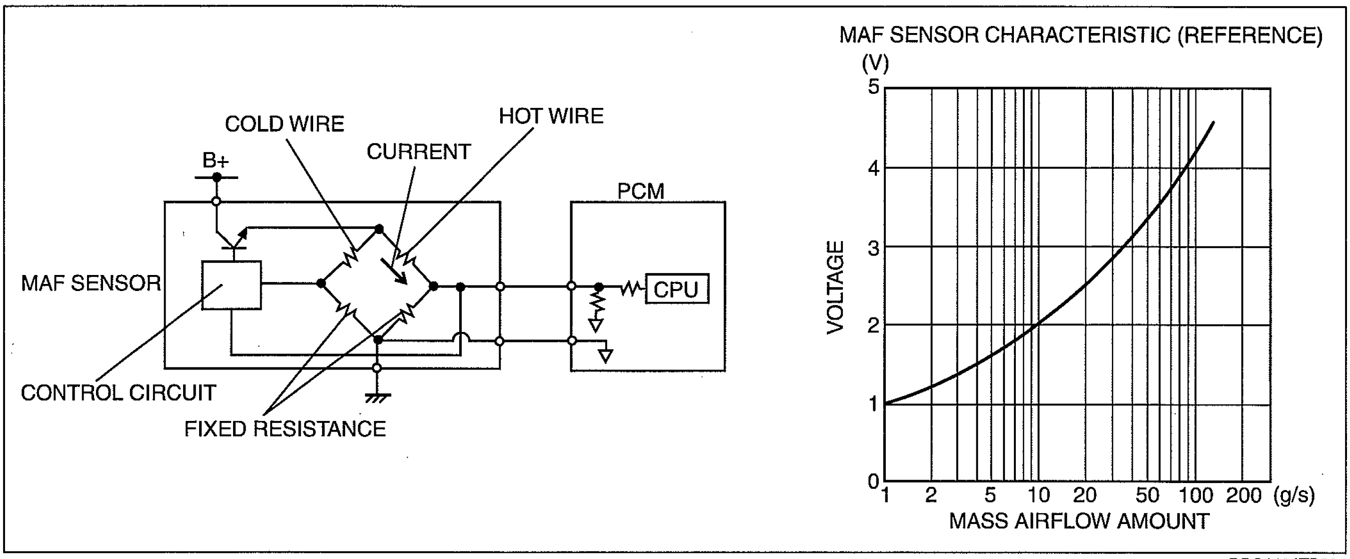

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INTAKE AIR TEMPERATURE (IAT) SENSOR NO.2 FUNCTION [WL-C, WE-C]

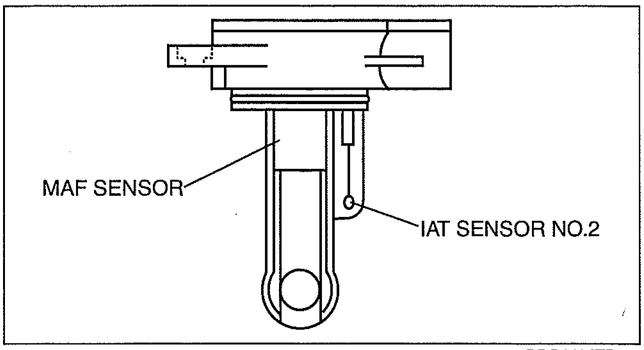
dcf014018842t05

- Detects the intake air temperature, and sends it to the PCM as an intake air temperature signal.

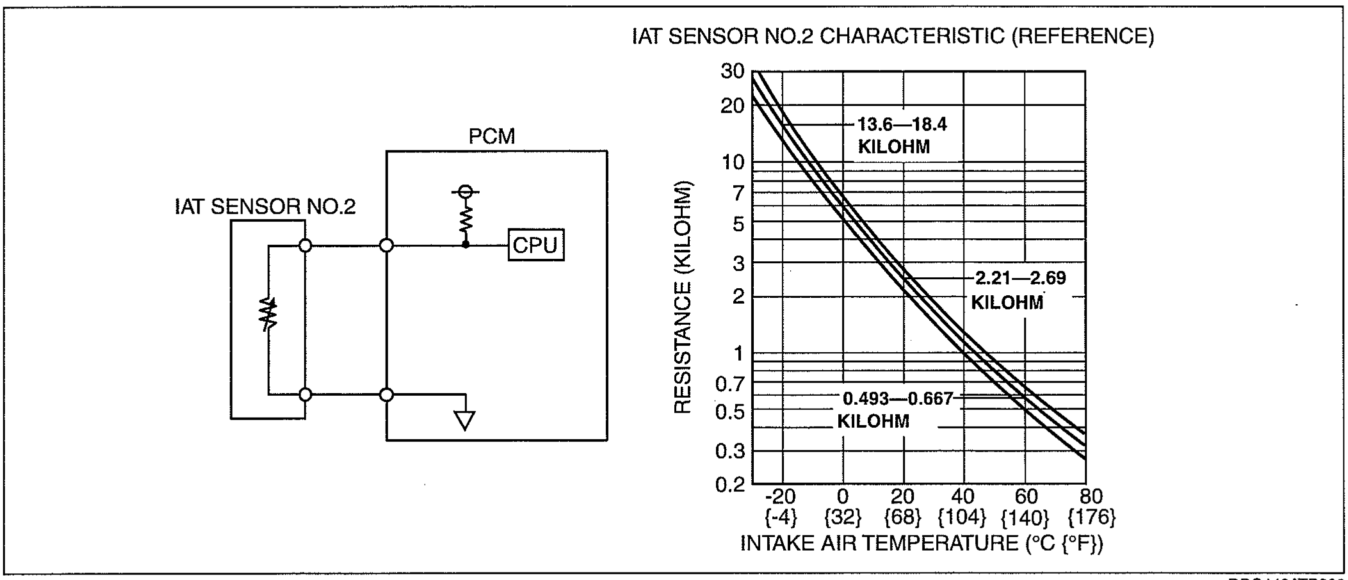
INTAKE AIR TEMPERATURE (IAT) SENSOR NO.2 CONSTRUCTION/OPERATION [WL-C, WE-C]

dcf014018842t06

- Installed to the air cleaner.
- Built into the mass airflow sensor.
- The thermistor type IAT sensor No.2 changes resistance according to the intake air temperature.
- The resistance decreases if the intake air temperature increases and, conversely, increases if the intake air temperature decreases.



DBG140ATB305



DBG140ATB306

INTAKE AIR TEMPERATURE (IAT) SENSOR NO.1 FUNCTION [WL-C, WE-C]

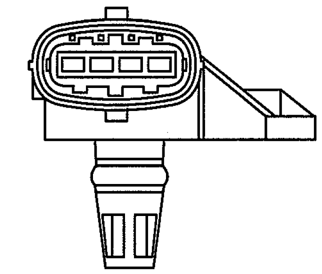
dcf014018842t07

- The IAT sensor No.1 detects the intake air temperature which has passed through the charge air cooler, and sends it to the PCM as an intake air temperature signal.

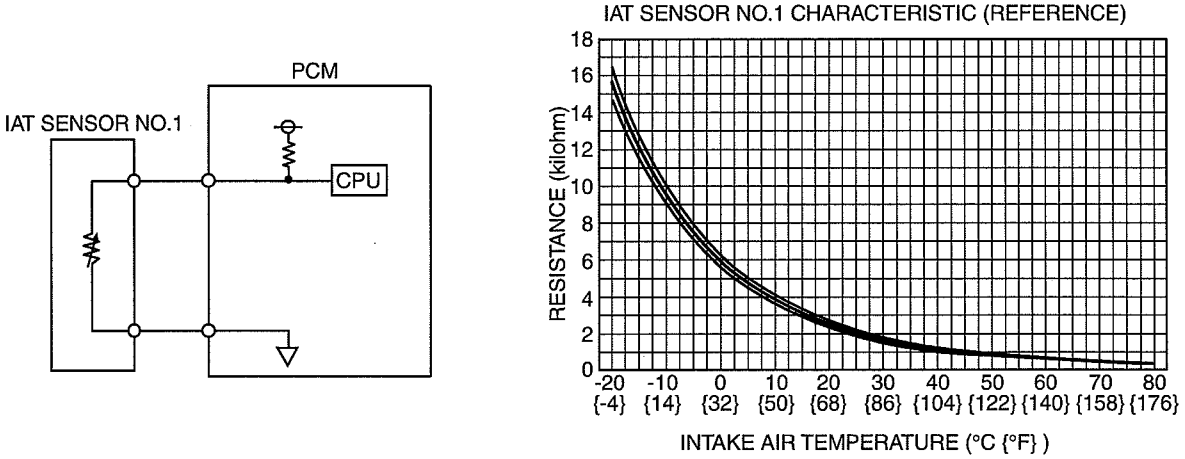
INTAKE AIR TEMPERATURE (IAT) SENSOR NO.1 CONSTRUCTION/OPERATION [WL-C, WE-C]

dcf014018842t08

- Built into the boost sensor.
- The boost sensor is installed to the intake manifold.
- The thermistor type IAT sensor No.1 changes resistance according to the intake air temperature.
- The resistance decreases if the intake air temperature increases and, conversely, increases if the intake air temperature decreases.



DBG140BTB551



DBG140BTB302

BOOST SENSOR FUNCTION [WL-C, WE-C]

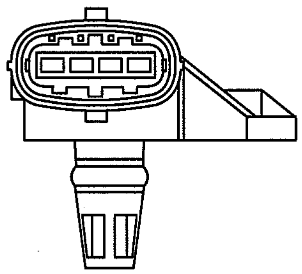
dcf014018212t03

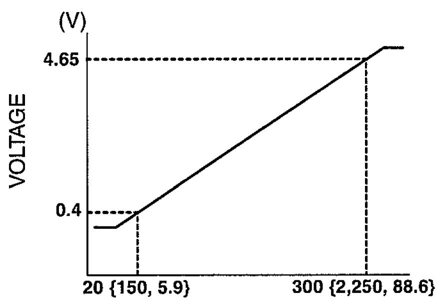
- The boost sensor detects the intake air pressure as an absolute pressure, and sends it to the PCM as an intake air pressure signal.

BOOST SENSOR CONSTRUCTION/OPERATION [WL-C, WE-C]

dcf014018212t04

- The boost sensor is installed to the intake manifold.
- The boost sensor is filled with crystal (silicon) and it is the semi-conductor pressure sensor which utilizes the characteristic of the electrical resistance that changes when the crystal is pressurized.
  *Absolute pressure is the pressure when vacuum is set as **0 kPa {0 mmHg, 0 inHg}**.
- With built-in IAT sensor No.2



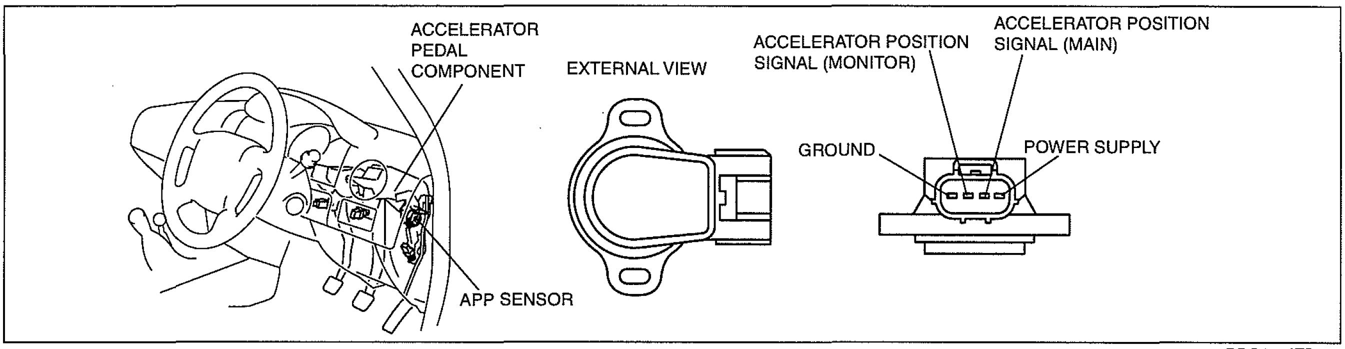


DBG140BTB301

ACCELERATOR PEDAL POSITION (APP) SENSOR FUNCTION [WL-C, WE-C]

dcf014041609t03

- The APP sensor is installed on the accelerator pedal, and detects how much the accelerator pedal is being depressed from the change in the resistance value (variable resistance).

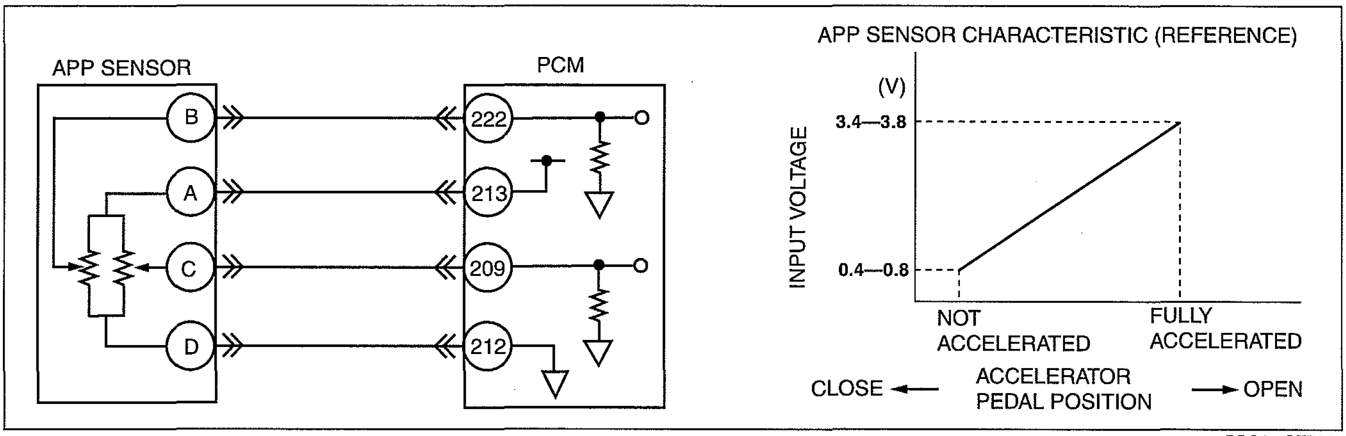


DBG140ATB459

ACCELERATOR PEDAL POSITION (APP) SENSOR DESCRIPTION [WL-C, WE-C]

dcf014041609t04

- The APP sensor is a potentiometer type and works is the same way as the throttle position sensor.
- The APP sensor contains two circuits for detecting accelerator position: main circuit and monitor circuit. The main circuit is used for the controls operated by the PCM (e.g. fuel injection amount control). The monitor circuit is used for detecting malfunctions in the APP sensor.
- When voltage difference between the main and monitor circuits increases, the PCM determines that the APP sensor is malfunctioning and stores DTC. Thus, a detection condition item has been added to DTC.
- The input voltage characteristic of the APP sensor is as shown.

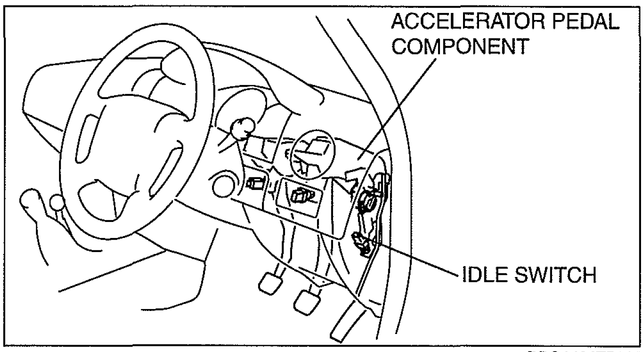


DBG140BTB307

IDLE SWITCH FUNCTION [WL-C, WE-C]

dcf014066470t03

- The idle switch is mounted to the accelerator pedal and inputs to the PCM whether the accelerator pedal is depressed or released. The PCM detects the idle switch signal as an idle signal.
- The idle signal inputted to the PCM is compared with the accelerator opening signal detected from the APP sensor, and on-board diagnostic of the idle switch and the APP sensor is performed.

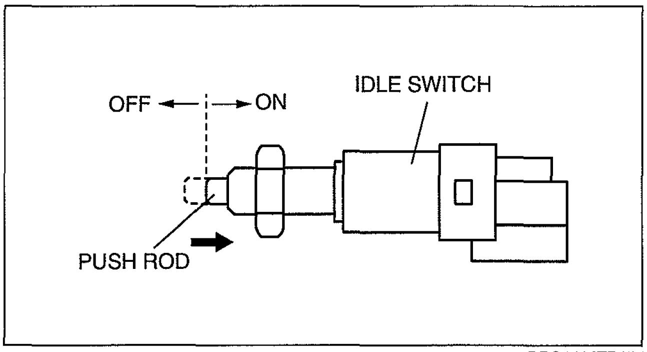


DBG140ATB002

IDLE SWITCH DESCRIPTION [WL-C, WE-C]

dcf014066470t04

- When the push rod is pushed in (the accelerator pedal is released) a preset amount, the contact of the idle switch closes, and the PCM determines that the engine is idling. When the contact of the idle switch is open (OFF), the PCM determines that the engine is not idling.



DBG140ATB455

FUEL TEMPERATURE SENSOR FUNCTION [WL-C, WE-C]

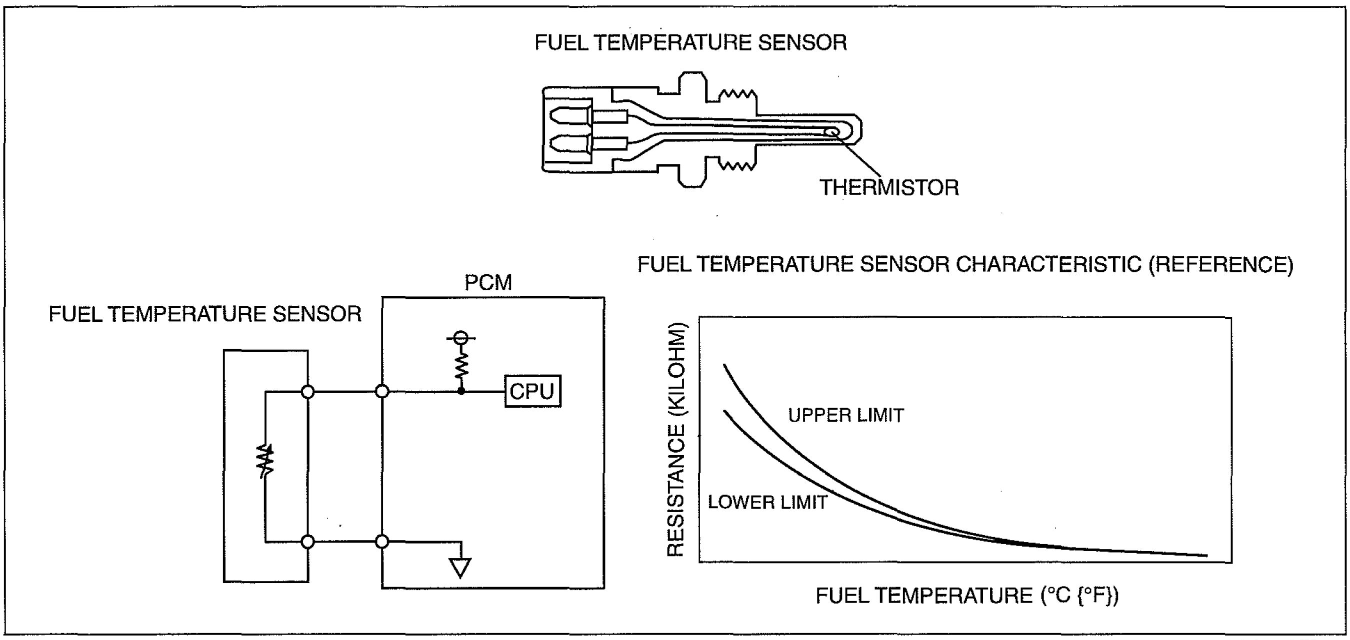
dcf014018843t03

- The fuel temperature sensor detects fluctuations in resistance of the thermistor to detect the fuel temperature which is necessary for the fuel injection amount correction.

FUEL TEMPERATURE SENSOR CONSTRUCTION/OPERATION [WL-C, WE-C]

dcf014018843t04

- Installed to the supply pump.
- The thermistor type fuel temperature sensor changes resistance according to the fuel temperature.
- The resistance decreases if the fuel temperature increases and, conversely, increases if the fuel temperature decreases.



DBG140BTB308

FUEL PRESSURE SENSOR FUNCTION [WL-C, WE-C]

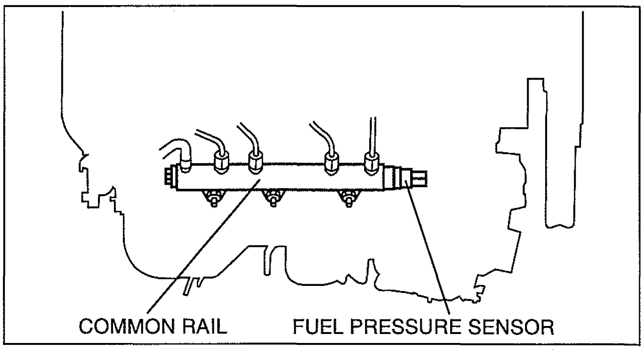
dcf014018213t01

- Measures the fuel pressure in the common rail.
- Measured fuel pressure is used for the fuel pressure control.

FUEL PRESSURE SENSOR CONSTRUCTION/OPERATION [WL-C, WE-C]

dcf014018213t02

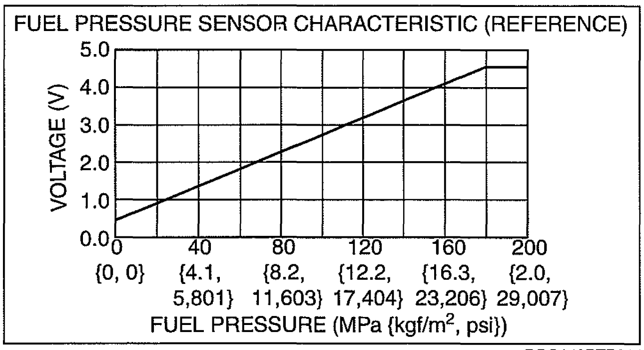
- Installed on the common rail.



DBG140BTB303

01

- Converts the fuel pressure in the common rail to a voltage signal and outputs it to the PCM.



DBG140BTB304

ENGINE COOLANT TEMPERATURE (ECT) SENSOR FUNCTION [WL-C, WE-C]

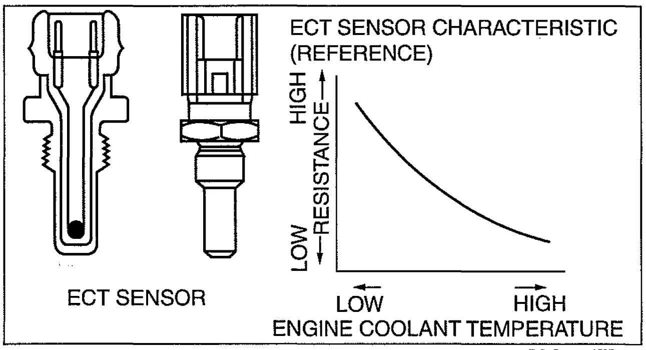
dcf014018841t03

- The ECT sensor detects the engine coolant temperature.

ENGINE COOLANT TEMPERATURE (ECT) SENSOR CONSTRUCTION/OPERATION [WL-C, WE-C]

dcf014018841t04

- The ECT sensor is thermistor type, and is installed in the cylinder head.
- The ECT sensor inputs the thermistor resistance, which changes according to the engine coolant temperature, to the PCM as a voltage.



DBG140ATB302

CAMSHAFT POSITION (CMP) SENSOR FUNCTION [WL-C, WE-C]

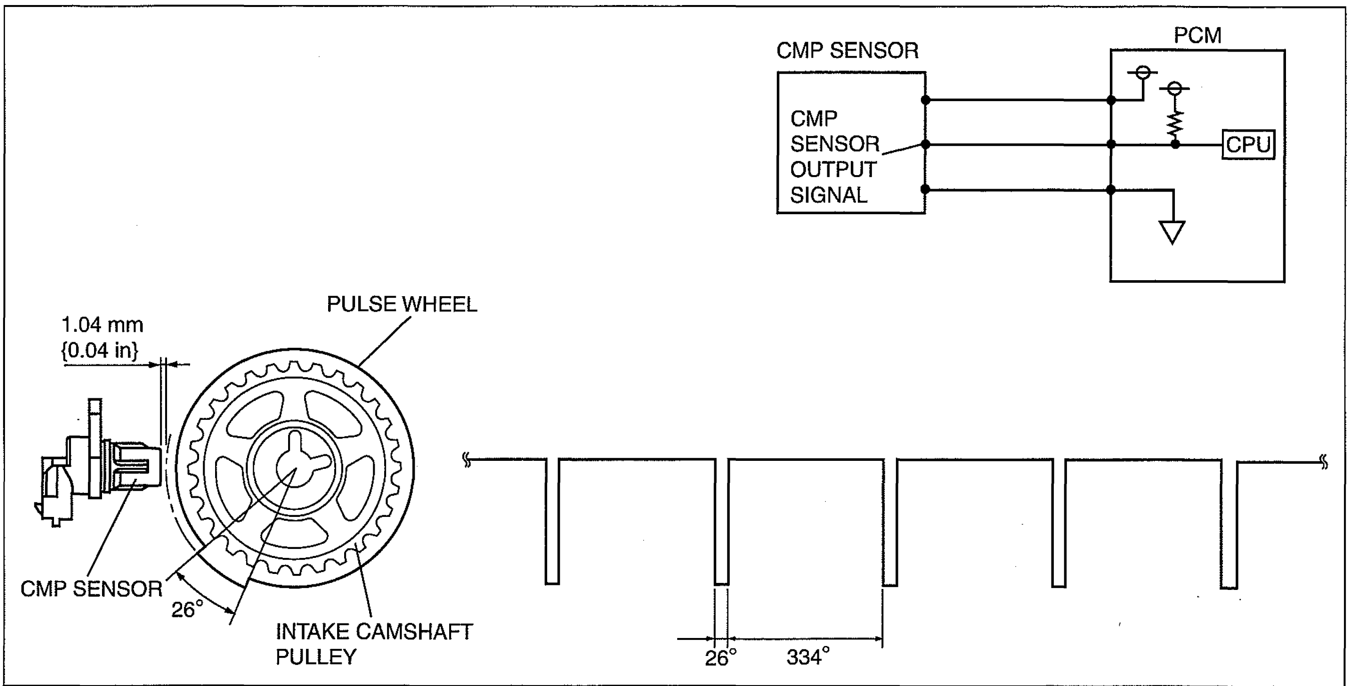
dcf014018230t01

- The CMP sensor detects pulse wheel of intake camshaft pulley rotational signals as cam angle signals.

CAMSHAFT POSITION (CMP) SENSOR CONSTRUCTION/OPERATION [WL-C, WE-C]

dcf014018230t02

- The CMP sensor is installed on the cylinder head on the front side of engine.
- The CMP sensor consists of a Hall element with a magnetic sensor, and a processing circuit that performs signal amplification and identification.
- CMP sensor is a hall-element type and it detects hall voltage generated by the change in magnetic flux of the magnet inside the sensor and hall element as the camshaft angle signal.
- One pulse are detected per one rotation of the camshaft by the projections on the intake camshaft pulley.
- If the CMP sensor is removed/installed or replaced, magnetized objects such as metal shavings adhering to the sensor could cause fluctuation in the magnetic flux of the magneto resistance element, causing abnormal sensor output which could adversely affect engine control.



DBG140BTB312

CRANKSHAFT POSITION (CKP) SENSOR FUNCTION [WL-C, WE-C]

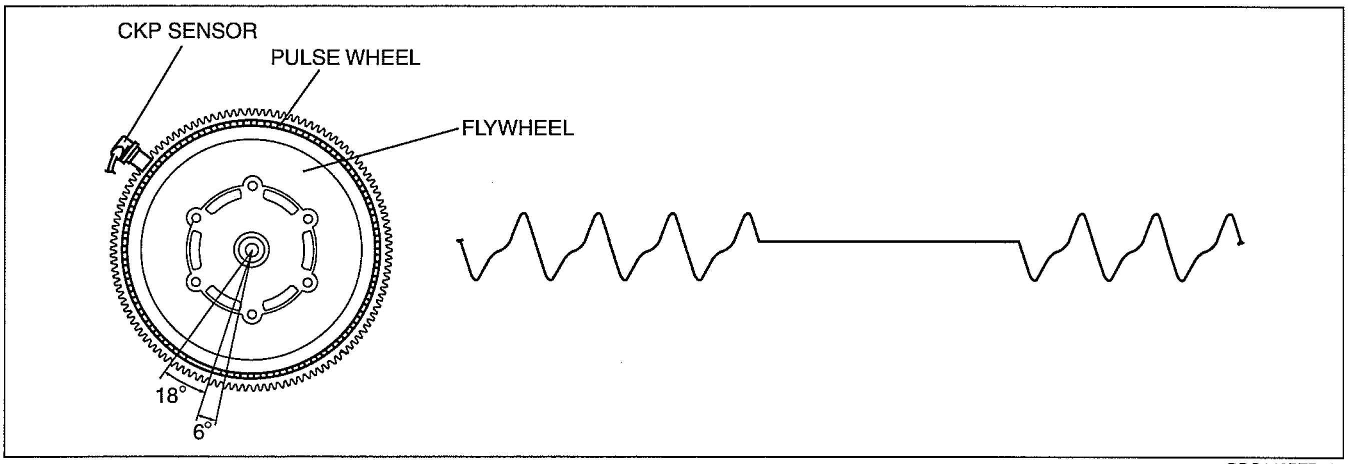
dcf014018220t03

- The CKP sensor detects pulse wheel rotational signals as crank angle signals.

CRANKSHAFT POSITION (CKP) SENSOR CONSTRUCTION/OPERATION [WL-C, WE-C]

dcf014018220t04

- The CKP sensor is installed to the clutch housing.
- The pulse wheel which is installed to the dual-mass flywheel has 58 projections and spaces with 6° of crank angle between each projection.
- The fluctuation in magnetic flux density detected by the magnetic pickup coil in the crankshaft position sensor is input to the PCM as a sensor output signal.
- If the crankshaft position sensor is removed/installed or replaced, magnetized objects such as metal shavings adhering to the sensor could cause fluctuation in the magnetic flux of the magnetic pickup coil, causing abnormal sensor output which could adversely affect engine control.

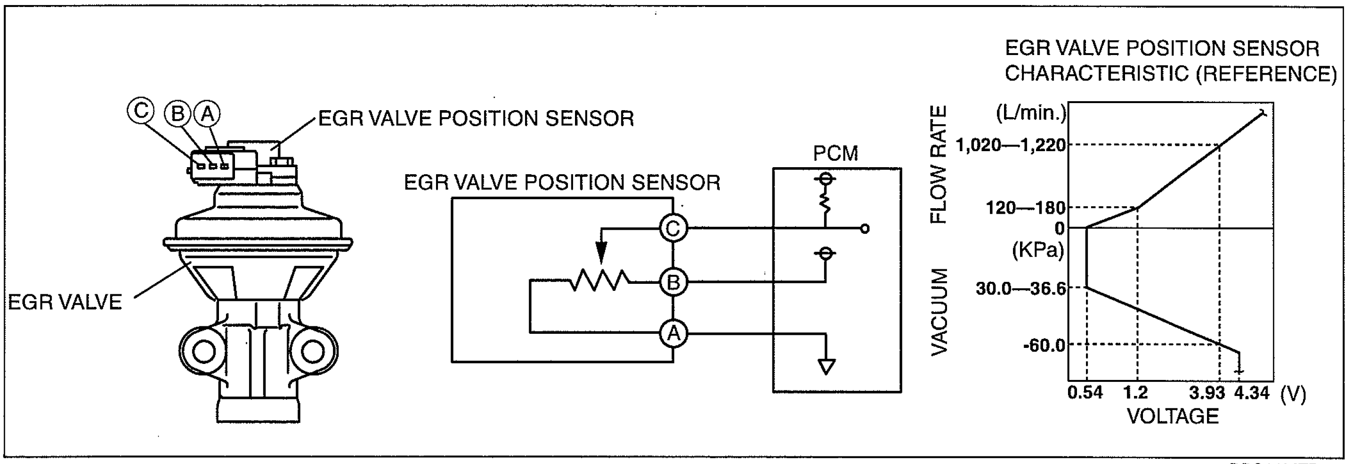


DBG140BTB313

EGR VALVE POSITION SENSOR FUNCTION [WL-C, WE-C]

dcf014020300t02

- The EGR valve position sensor measures EGR valve lift amount and sends signal to the PCM, allowing the PCM to diagnose a EGR system malfunction when there is a discrepancy between target and actual opening angles of the EGR valve.



DBG140ATB311

CLUTCH PEDAL POSITION (CPP) SWITCH FUNCTION [WL-C, WE-C]

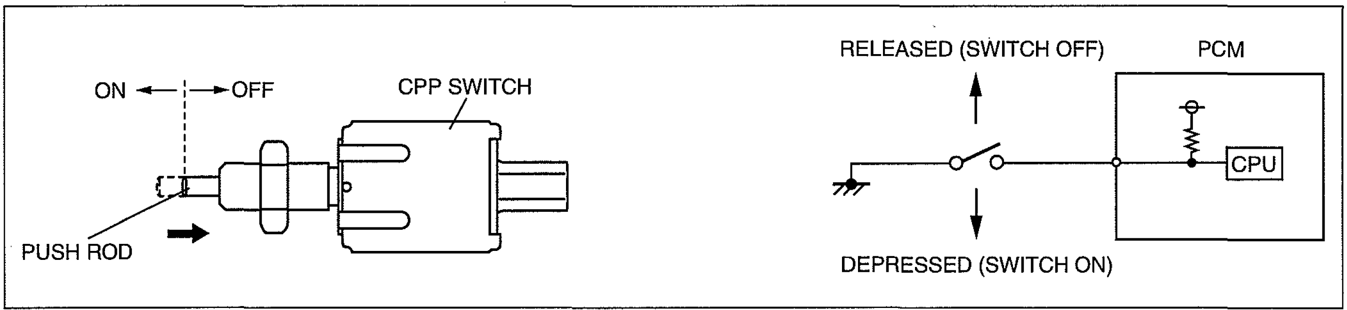
dcf014018660t03

- The CPP switch detects the clutch connection condition.

CLUTCH PEDAL POSITION (CPP) SWITCH DESCRIPTION [WL-C, WE-C]

dcf014018660t04

- The CPP switch is mounted to the top of the clutch pedal. When the clutch pedal is depressed, the contact of the CPP switch closes (ON) and the PCM detects below 1.0V. When the clutch pedal is released, the contact opens (OFF) and the PCM detects B+.



DBG140BTB550

NEUTRAL SWITCH FUNCTION [WL-C, WE-C]

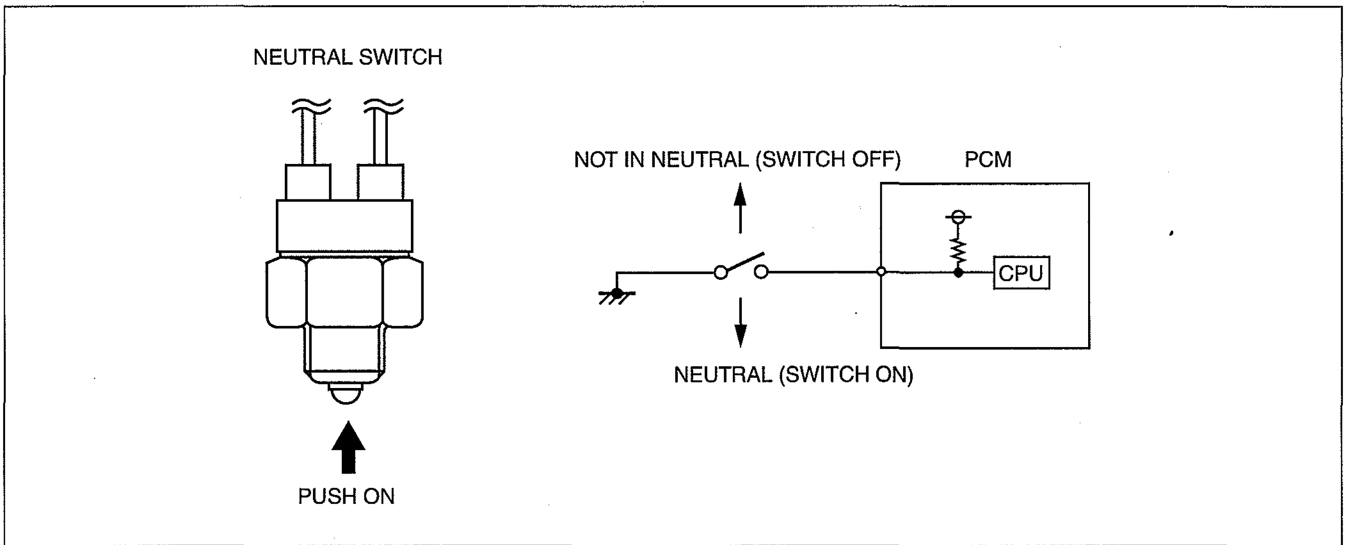
dcf014017640t03

- Detects shift lever neutral position.

NEUTRAL SWITCH CONSTRUCTION/OPERATION [WL-C, WE-C]

dcf014017640t04

- While the shift lever is in the neutral position, the contact point closes (ON) and the PCM detects below 1.0 V. While the shift lever is not in the neutral position, the contact point opens (OFF) and the PCM detects B+.



DBG140BTB305